



PE CIVIL-STRUCTURAL EXAM 05

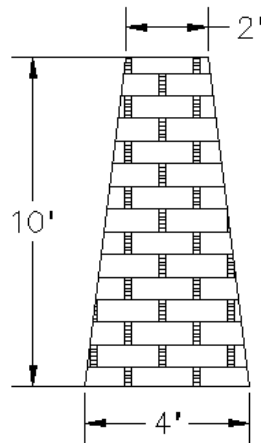
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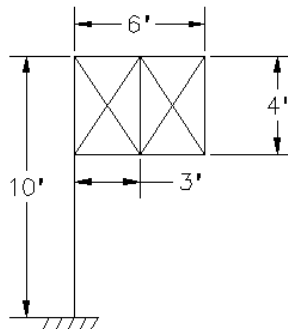
Problems

[Click here for Solutions](#)

1. A trapezoidal stone masonry abutment is constructed to support a culvert. The abutment is 12 ft in length and the cross section of the abutment is as shown below. What is most nearly the self-weight of the abutment?

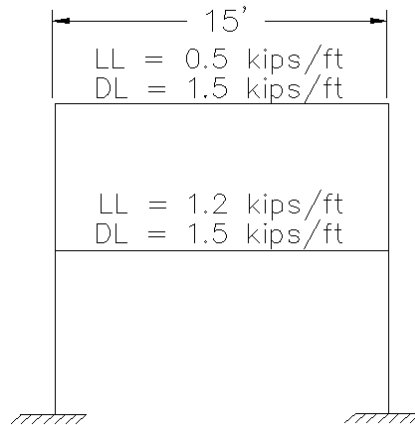


- (A) 35 kips
(B) 61 kips
(C) 76 kips
(D) 92 kips
2. A cantilevered traffic light pole is designed as shown below. The vertical and horizontal members are fabricated with HSS 6X6X1/2 and the diagonal members are fabricated with HSS 2X2X1/4. The whole assembly is to be erected at the job site. What is the capacity of the hoisting mechanism required to lift the framing assembly? Assume additional rigging weight of 10% of the framing assembly.



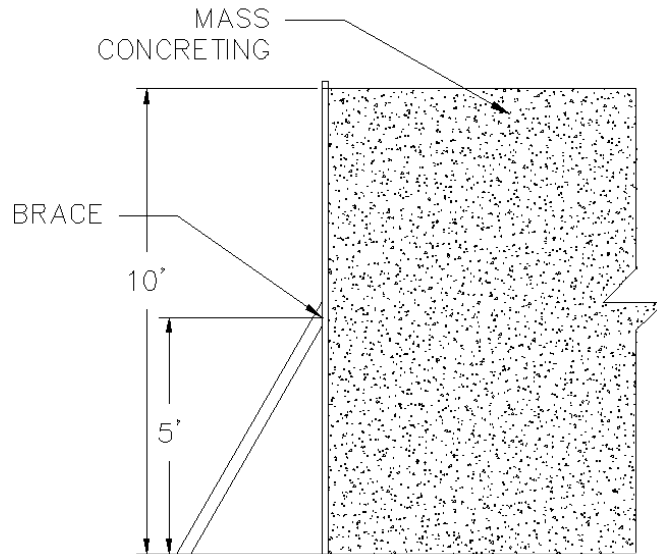
- (A) 1165 lb
(B) 1282 lb
(C) 1382 lb
(D) 1400 lb

3. The frame shown below belongs to the bay of a building. The columns shown are exterior columns. Tributary area for each column is 112.5 ft^2 . What is the total live load acting on the base of a ground floor column? Apply all reduction factors as applicable.



- (A) 6 kips
(B) 12.75 kips
(C) 12.2 kips
(D) 4.4 kips
4. For a sloped roof, what is the basis for applying slope adjustment factors to reduce the live loads per the provisions of ASCE7-16 and IBC 2018?
- (A) Structural capacity of roofing members increases with inclination
(B) Seismic load governs for sloped roof and live loads can be neglected
(C) Sloped roofs have reduced accessibility and are less likely to be occupied
(D) Snow load accumulation is lesser in sloped roofs

5. In a mass concreting job, fresh concrete is poured to cast an embankment. The first pour is 10 ft high. The formwork boards are braced with a 4X4 and are prevented from sliding at the base. The boards are braced at mid height as shown in the sketch below. What is the lateral force exerted by the freshly poured concrete at the point of bracing? Unit weight of fresh concrete = 150 pcf



- (A) 5 kips/ft
(B) 7 kips/ft
(C) 10 kips/ft
(D) 12 kips/ft
6. A 10 story hospital is to be built in New York. The infrastructure is classified as an essential structure to the county. The site is flat and topographic effects can be neglected. There are numerous closely spaced commercial houses and up to 2500 ft from the site upto which the area can be classified as suburban. Beyond that, there is open terrain with scattered obstructions. What is the wind exposure category to be used in the design of the building?
- (A) Exposure A
(B) Exposure B
(C) Exposure C
(D) Exposure D

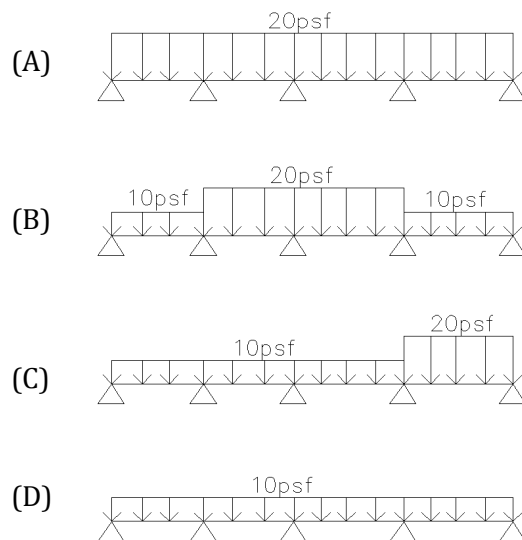
7. Calculate the seismic demand on a cantilevered unbraced parapet wall located at the roof of a building. The importance factor of the component can be taken as 1. The weight of the wall is 50 kips and the design short period spectral acceleration is 0.5g.

(A) 25 kips
(B) 30 kips
(C) 40 kips
(D) 50 kips

8. According to AASHTO LRFD specifications, to calculate the total design vehicular live load, which one of the following must be considered?

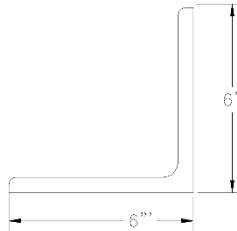
(A) Design lane load and Design truck load,
(B) Tandem axle of 25 kip each spaced 4.0 ft apart,
(C) Design truck load plus design tandem axle load plus design lane load,
(D) Design truck or Design tandem plus 0.64 klf of uniformly distributed lane load

9. In addition to uniform balanced snow load, ASCE 7-16 requires that designs incorporate partial loadings and unbalanced snow loads. For a roofing system with 4 spans, the full uniform snow load is 20psf. Which of the following loading conditions is not recommended per ASCE 7-16 provisions?



- 10.** Calculate the ice load due to weight of ice formed on the members of a mounting frame at the roof of a 10 story building that is 100ft tall. The frame is constructed using L6X6X1/2 members. The building is classified as risk category III. The nominal ice thickness formed by freezing rain at a height of 33 ft in the area is 1.5 inches. Topographic effects can be neglected for the area.

- (A) 24.6 lb/ft
- (B) 28.3 lb/ft
- (C) 30.1 lb/ft
- (D) 32.7 lb/ft



- 11.** A structural designer wants to account for impact loads in a semi-commercial building. Which one of the following provisions is not true per ASCE 7-16?

- (A) Loads due to unusual vibrations should be accounted for while designing machine foundations. These load are not provided in ASCE 7-16 and should be provided by the equipment manufacturer,
- (B) The operating weights of motor driven light machinery should be increased by 20%,
- (C) The vertical impact due to wheel loads of pendant operated bridge crane is taken as 10% of the wheel load,
- (D) The equipment/moving load factored with appropriate impact factors need not have the additional 1.6 live load factor for LRFD design.

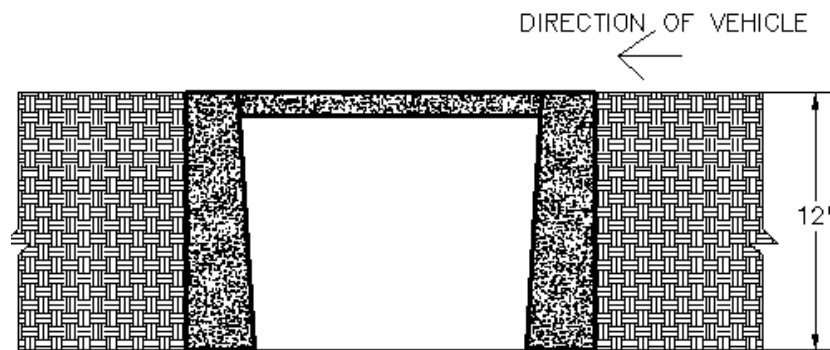
12. The height of an abutment wall supporting a bridge deck is 12ft. The abutment has backfill on one side as shown below. What is the additional lateral load due to vehicular load on top of the embankment? Calculate live load surcharge per AASHTO LRFD provisions.

Active earth pressure coefficient = 0.33

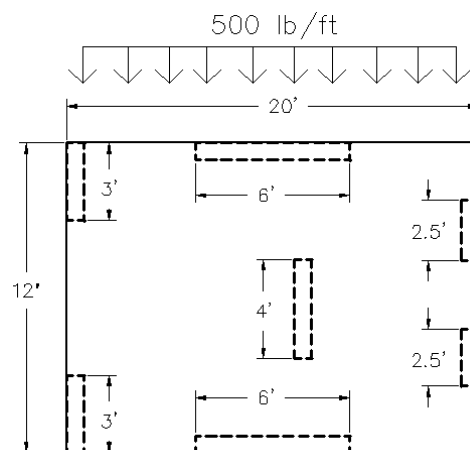
Earth pressure coefficient at rest = 0.5

Passive earth pressure coefficient = 3

Density of soil, $\gamma = 110$ pcf



- (A) 1288 lb/ft
(B) 1848 lb/ft
(C) 2048 lb/ft
(D) 2288 lb/ft
13. The plan shown below is a reinforced concrete floor and is supported by shear walls underneath. The in plane stiffness of the walls supporting the floor is directly proportional to the length of the wall. Out of plane stiffness can be neglected. The center of mass and center of eccentricity coincides with each other and any accidental eccentricity can be neglected. What is the shear load acting on the center wall?



- (A) 2000 lbs
(B) 2666.67 lbs
(C) 3000 lbs
(D) 3333.33 lbs

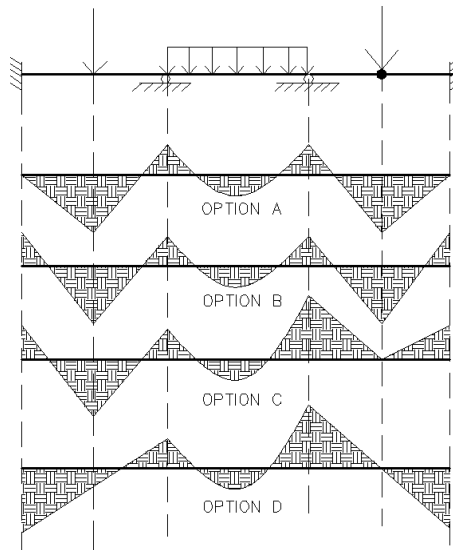
14. The maximum unfactored axial loads on a column supporting a roof per ASCE 7-16 are as follows:

- a. Dead Load = 25 kips
- b. Load from an equipment = 5 kips
- c. Roof live load = 10 kips
- d. Snow load = 8 kips
- e. Wind load = 15 kips

What is the governing factored load for strength design?

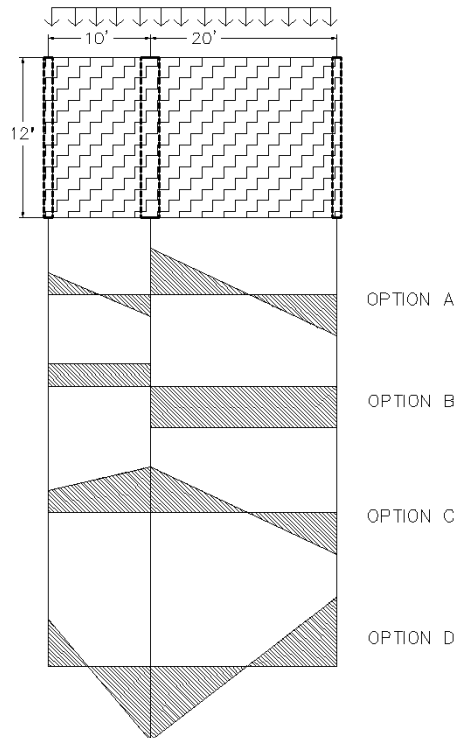
- (A) 41.5 kips
- (B) 50 kips
- (C) 55 kips
- (D) 62 kips

15. For the given three span beam with a hinge at the center of the third span, what is the correct bending moment diagram?



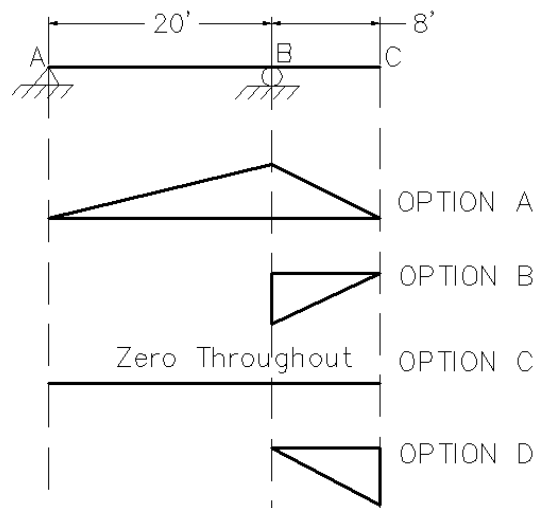
- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

16. A flexible diaphragm is supported by three walls. The center wall is twice as stiff as the outer walls. Select the correct shear force diagram for the diaphragm.



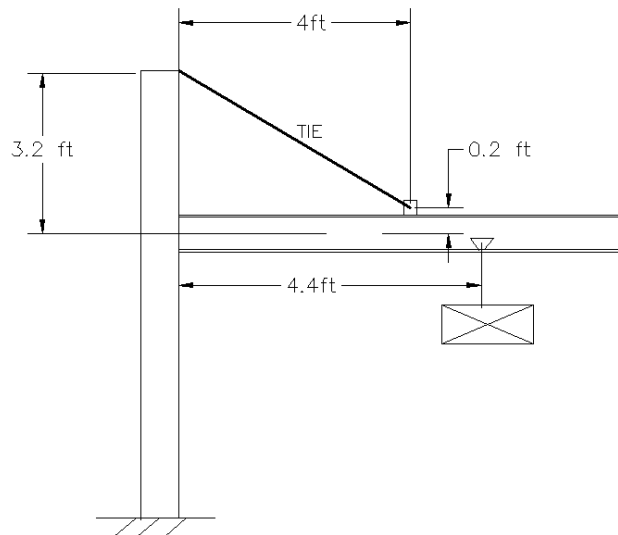
- (A) Option A
(B) Option B
(C) Option C
(D) Option D

17. For the overhanging beam shown below, select the correct influence line diagram for maximum moment at B, when a concentrated load passes from point A to point C

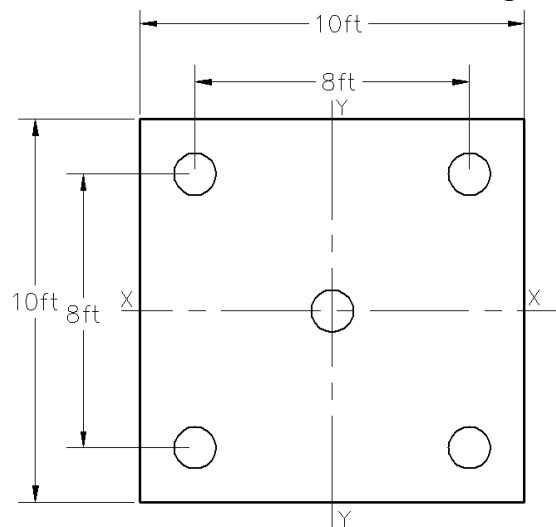


- (A) Option A
(B) Option B
(C) Option C
(D) Option D

18. For the configuration below, the weight of the pick is 500 lbs. The tie is offset from the center of the beam by 0.2 ft. What is the maximum tension in the tie? Self weight of the jib can be neglected.

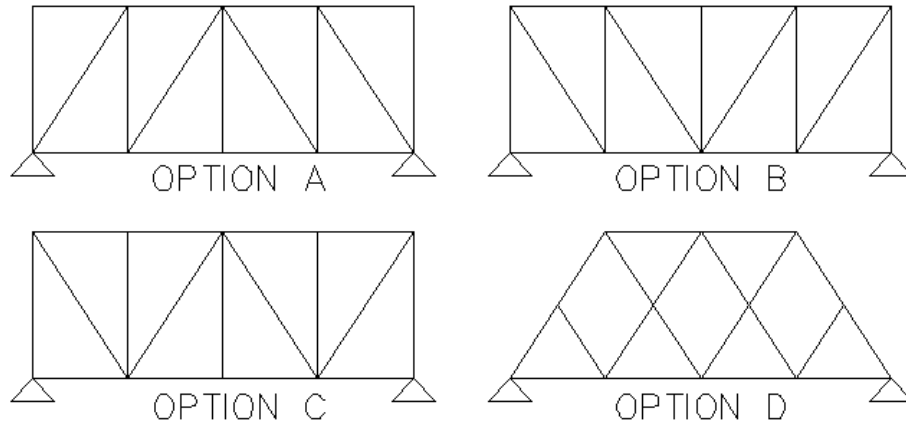


- (A) 820 lbs
(B) 860 lbs
(C) 900 lbs
(D) 920 lbs
19. The plan below shows a rigid concrete pile cap with 5 piles underneath. The maximum axial load acting on the pile cap is 2000 kips and the maximum moments are $M_{xx} = 500 \text{ kip}\cdot\text{ft}$ and $M_{yy} = 300 \text{ kip}\cdot\text{ft}$. What is the maximum axial load acting on an individual pile?



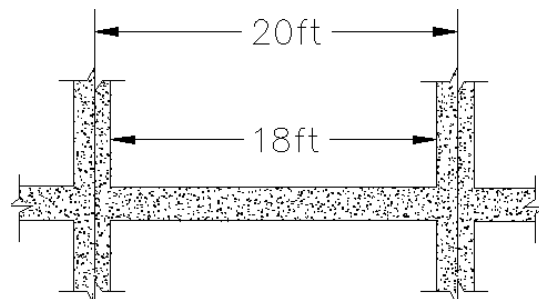
- (A) 400 kips
(B) 450 kips
(C) 500 kips
(D) 600 kips

20. Which of the following truss models have only compressive forces in all diagonal members ?



- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

21. Partial elevation of an interior reinforced concrete beam is shown below. The frame is designed and analyzed as a special moment resisting frame. During sidesway, the maximum probable flexural strengths at the left and right ends are 500 kip*ft and 600 kip*ft. The unfactored uniform dead load acting on the beam is 2 kip/ft. The unfactored live load acting on the beam is 1 kip/ft. What is the factored design shear to be considered for the beams?

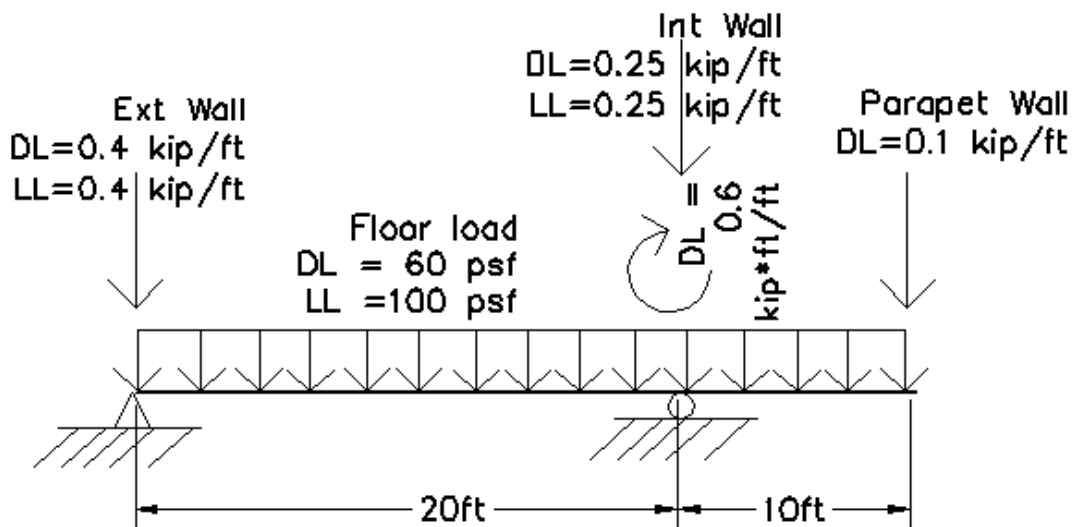


- (A) 36 kips
- (B) 60 kips
- (C) 92 kips
- (D) 105 kips

22. A beam with a rectangular cross section is subjected to uniformly varying load and multiple point loads along the span. Which of the following best describes the distribution of shear stress in the section?

- (A) For a symmetrical rectangular section, the shear stress is uniform along the height of the section,
- (B) Shear stresses are maximum at the top and bottom surfaces and minimum at the neutral axis,
- (C) Shear stresses are maximum at the neutral axis and zero at the top and bottom of the section,
- (D) Shear stress varies linearly from top to bottom of the section.

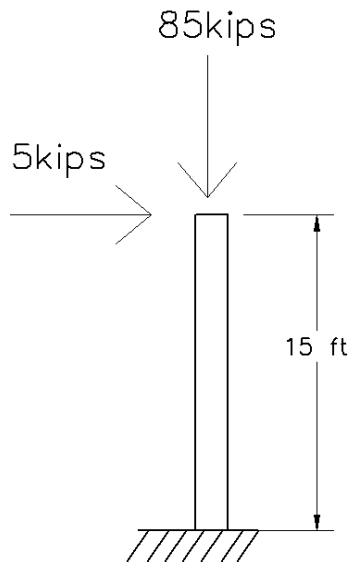
23. Loads acting on a simply supported floor beam is shown below. The floor beams are spaced 8 ft c/c. What is the maximum service load shear acting on the beam ?



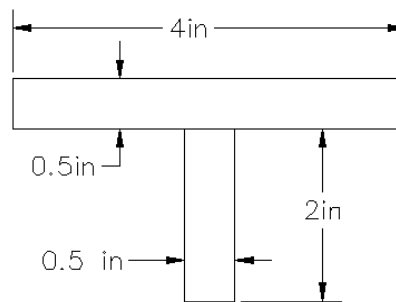
- (A) 2.08 kips
- (B) 8.96 kips
- (C) 15 kips
- (D) 16.64 kips

24. A cantilevered column is subjected to 5 kip lateral load at the top and an 85 kips concentric axial load. The lateral stiffness of the column is 2.5 kips/in. What is the total moment acting at the base of the column?

- (A) 60 kip*ft
(B) 75 kip*ft
(C) 90 kip*ft
(D) 100 kip*ft

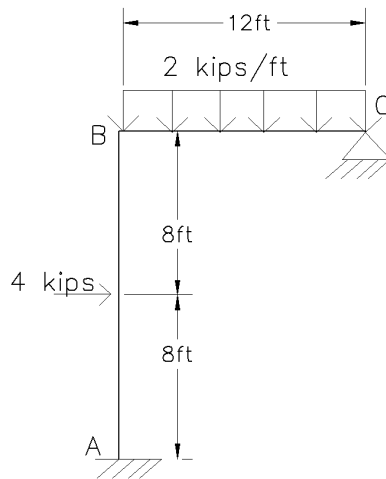


25. For the T section shown below, what is the location of the plastic neutral axis from the base of the section?



- (A) 1.5 in
(B) 1.833 in
(C) 2 in
(D) 2.125 in

26. For the framing configuration below, what is the moment reaction under column AB? The modulus of elasticity is constant. Moment of inertia of beam BC = 800 in^4 and column AB = 550 in^4 .



- (A) 1.2 kip*ft
 (B) 2.3 kip*ft
 (C) 3.5 kip*ft
 (D) 5.8 kip*ft
27. Which of the following biaxial states of stress represents a body in pure shear?

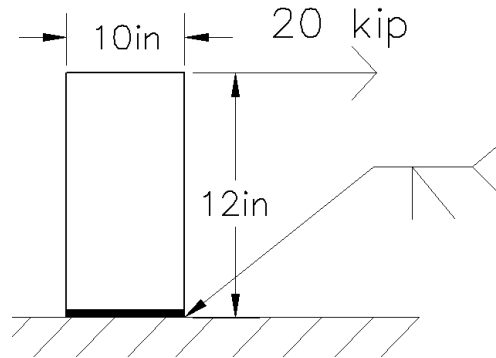
(A)
$$\begin{bmatrix} \sigma_1 & \tau_{xy} \\ \tau_{yx} & \sigma_2 \end{bmatrix}$$

(B)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

(C)
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & -\sigma_1 \end{bmatrix}$$

- (D) None of the above

28. A 10 in X 0.5 in X 12 in plate is rigidly connected using full penetration weld as shown in the figure below. The plate is subjected to a lateral load of 20 kip at the top. What is the resultant stress in the weld?



- (A) 2 kip/in
 (B) 5.63 kip/in
 (C) 7.81 kips/in
 (D) 14.54 kips/in
29. When a threaded rod is tightened with a wrench, which of the following stresses are acting in the assembly?
- I) Bending of rod due to wrench torque,
 - II) Bearing stresses under locking nuts and washers,
 - III) Tension in rod due to preload.
- (A) I, II and III,
 (B) II only,
 (C) I and II,
 (D) II and III

30. A lintel supports unreinforced masonry above it. The span of the lintel is 8ft. What is the maximum allowable deflection in the lintel under combined service dead and live loads?

- (A) 0.16 in
- (B) 0.26 in
- (C) 0.53 in
- (D) 1 in

31. A simply supported truss is shown below. A 30 kips load acts at joint B. The forces resulting from the load on each member is shown on each member (tension +ve). What is the maximum horizontal deflection on joint B due to the load? Area = 10 in^2 and $E = 29000 \text{ ksi}$ for all members.

Lengths of members

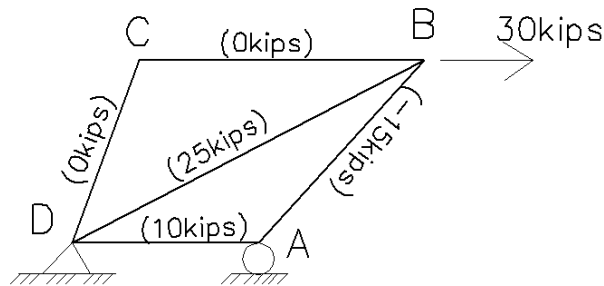
AB = 120 in

BC = 150 in

CD = 100 in

AD = 100 in

BD = 180 in



- (A) 0.011 in
- (B) 0.017 in
- (C) 0.023 in
- (D) 0.027 in

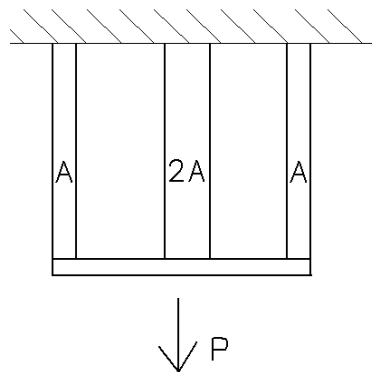
32. What is the deflection at the free end of the system? The center bar has two times the area of the edge bar. All three bars are equal in length.

$A = 10 \text{ in}^2$

$E = 29000 \text{ ksi}$

$L = 15 \text{ ft}$

$P = 200 \text{ kips}$



- (A) 0.03 in
- (B) 0.03 in
- (C) 0.04 in
- (D) 0.05 in

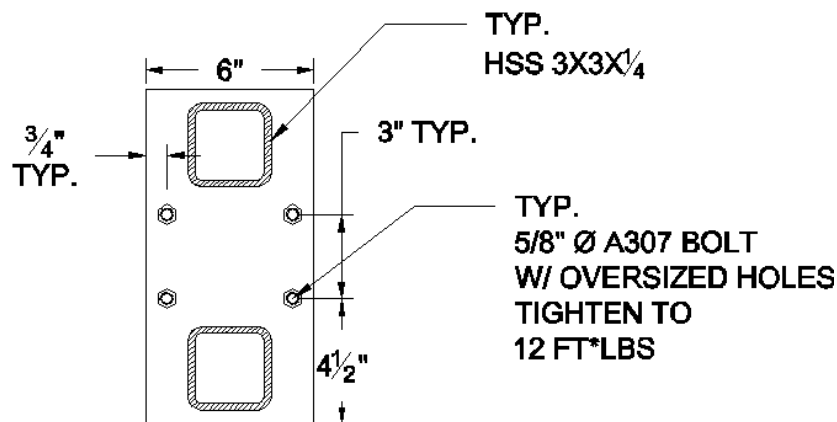
33. A circular shaft of diameter 8 inches and length 12 ft is fixed at both ends against deflections and rotations about all directions. It is subjected to a torsional moment of 500 kip*ft at the center. What is the angle of twist in the shaft? Modulus of elasticity, $E = 29000$ ksi and Poisson's ratio is 0.3.

- (A) 0.048 rad
- (B) 0.096 rad
- (C) 0.13 rad
- (D) 0.192 rad

34. A building is to be analyzed for progressive collapse after sudden failure of ground floor column. Which of the following analysis is not required for alternate load path analysis?

- (A) Static analysis with factored service loads,
- (B) Nonlinear analysis for large deformations,
- (C) Dynamic analysis simulating sudden removal of column,
- (D) Foundation uplift and bearing under failed column.

35. The figure below shows a 4 bolt connection detail. A special inspector needs to verify the bolting compliance on site. What is the maximum diameter of holes that he can approve?



- (A) 5/8 inch
- (B) 11/16 inch,
- (C) 13/16 inch.
- (D) 15/16 inch

36. For a concrete construction job site, which of the following can be inspected intermittently and does not require constant monitoring by the special inspector?

- (A) Verifying proper spacing, cover and size of rebars,
- (B) Adhesive anchors installation to fasten electrical panel unit on a shear wall cast two weeks ago,
- (C) Verifying groove welds used for rebar lapping,
- (D) Application of prestressing forces on a multi span concrete girder.

37. A shop drawing package submitted to the building engineer of record was returned to the contractor with remarks as follows:

WE ARE SENDING:	SUBMITTED FOR:	ACTION TAKEN:
☒ Shop Drawings	☒ Approval	☐ Approved as Submitted
☐ Letter	☐ Your Use	☒ Approved as Noted
☐ Prints	☐ As Requested	☐ Resubmit
☐ Change Order	☐ Review and Comment	☐ Submit
☐ Plans	☐ Other:	☐ Returned
☐ Samples	SENT VIA:	☐ Returned for Corrections
☐ Specifications	☒ Attached	☐ Due Date:
☐ Other:	☐ Separate Cover Via:	☐ Other:

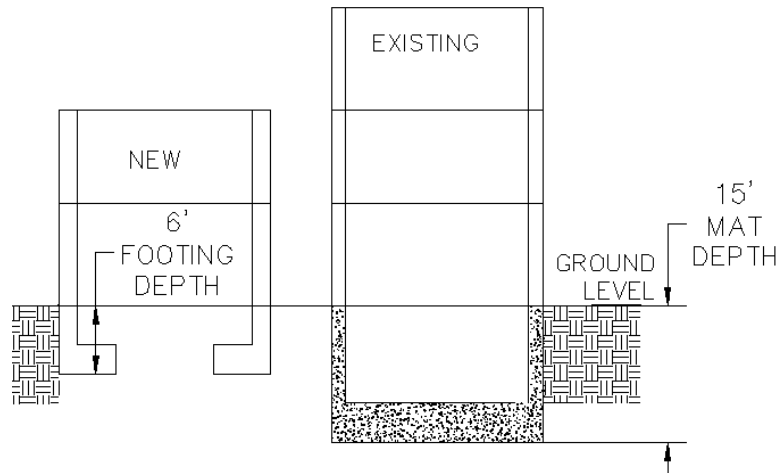
What does the action taken here typically mean?

- (A) Revisions are required before proceeding,
- (B) The contractor can proceed with field installation and the minor changes are acceptable,
- (C) Submittal package needs to be resubmitted,
- (D) Architect's approval is required before proceeding any further.

38. For a concrete construction project, what is the purpose of reshoring?

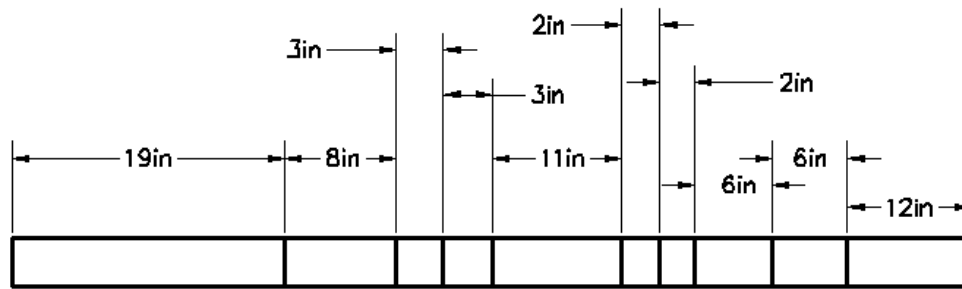
- (A) To support the weight of concrete beams and slabs during concrete placement,
- (B) To support the beams and slabs after stripping off formwork and redistribute loads safely to the base of the structure,
- (C) To control lateral displacements in the falsework system,
- (D) To help the concrete members gain compressive strength faster.

39. A two story building is to be constructed next to an existing building with a basement. The depth of foundation of the new building is 6 feet and the depth of basement mat of the existing building is 15 feet below grade. Assuming that the footing load due to the new structure follows a 45 degree influence line, what is the minimum distance between the two buildings so that the existing building is away from the influence of new building loads?



- (A) 6 ft
 - (B) 8 ft
 - (C) 9 ft
 - (D) 15 ft
40. Per IBC 2018, which of the following is not true about protecting adjoining existing facilities during construction activities ?
- (A) Owners of adjoining buildings should be notified via a written notice before any excavation work.
 - (B) Proper provisions for water runoff during construction or demolition should be made,
 - (C) Owners of adjacent buildings should be notified so that all tenants will vacate before a deep foundation excavation job in an adjacent lot with appropriate financial compensation.
 - (D) Protections should be provided for adjacent footings, roofs, chimneys, skylights and adjacent walls.

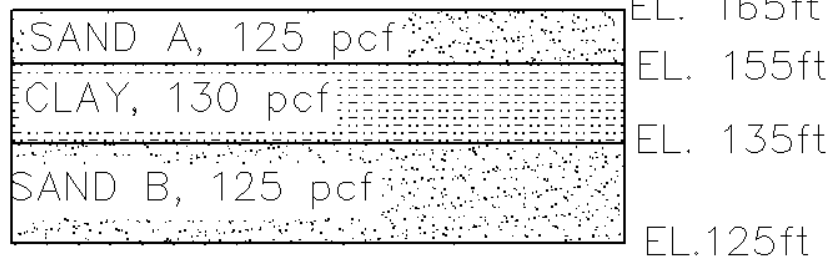
41. A 6ft core drilling was carried out at a site with bedrock and the core recovery is:



What is the quality of rock?

- (A) Fair
- (B) Good
- (C) Very Good
- (D) Excellent

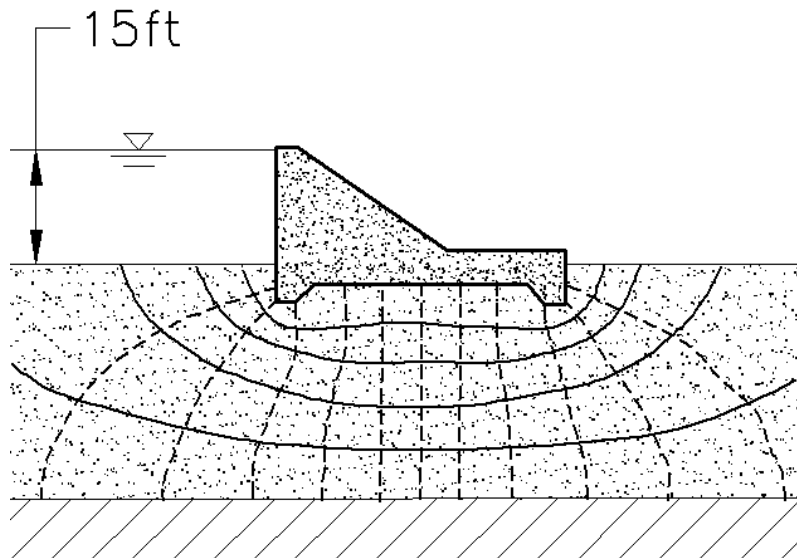
42. Clay layer is bound between two sand layers as shown below:



The coefficient of consolidation of the clay is $0.29 \text{ ft}^2/\text{day}$. Determine the time in days for 60% average degree of consolidation.

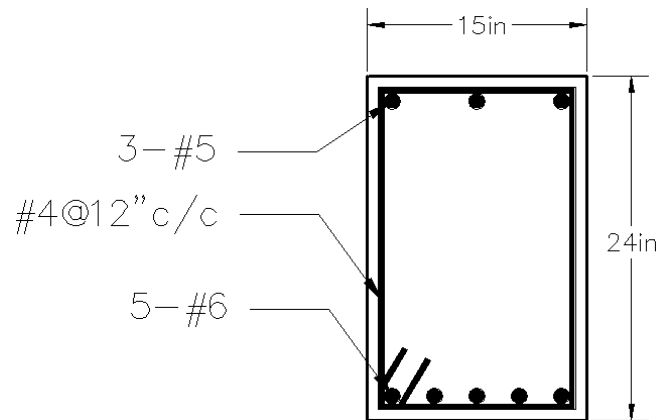
- (A) 59 days
- (B) 99 days
- (C) 180 days
- (D) 395 days

43. The figure below shows a flow net for seepage under an embankment. Under the flow net, there is an impervious layer. The permeability of the soil is 0.0001 ft/min. What is the estimated seepage per day?



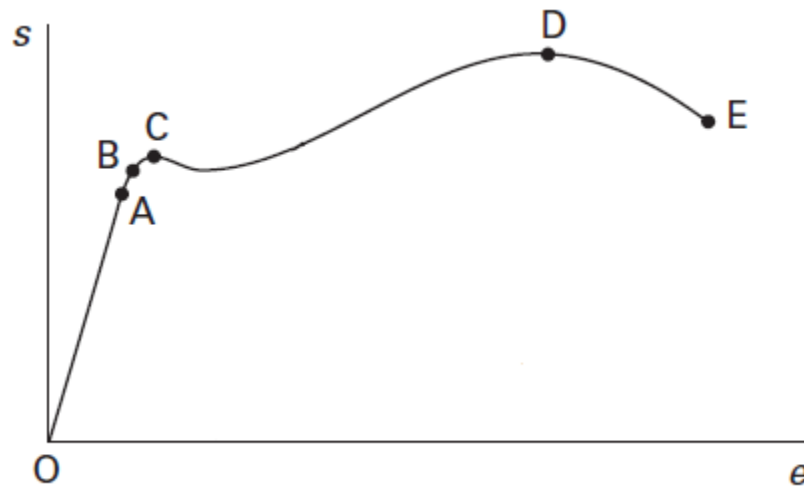
- (A) 0.9 cubic feet/ day/feet
 (B) 1.1 cubic feet/ day/feet
 (C) 1.3 cubic feet/ day/feet
 (D) 1.5 cubic feet/ day/feet
44. What is the modulus of rupture of a 4ksi lightweight plain concrete when it is subjected to bending?
- (A) 0.35 ksi
 (B) 0.4 ksi
 (C) 0.45 ksi
 (D) 0.5 ksi

45. The section of a simply supported RCC beam is shown below. If the immediate deflection of the beam is 1 inch, what is the estimated long term deflection after 8 years?
Effective cover can be assumed as 2 inches.



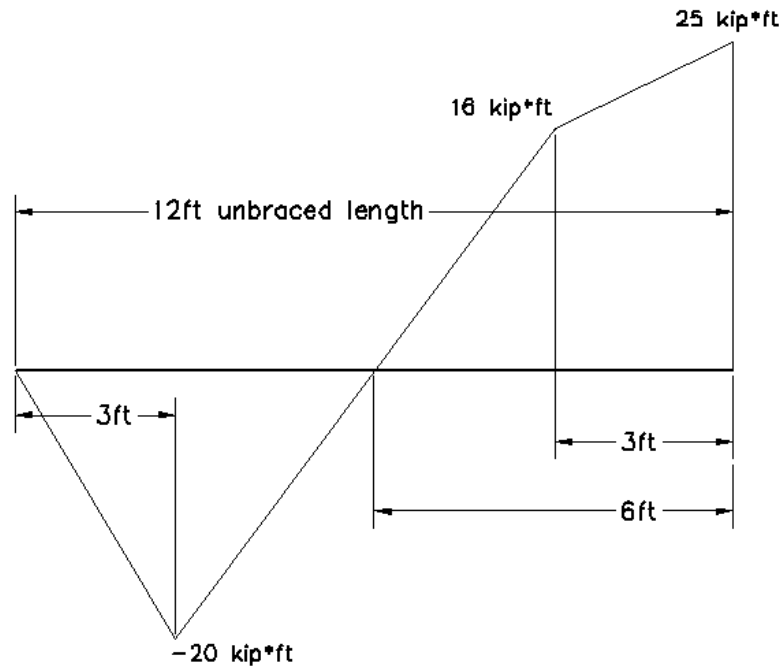
- (A) 1.25 in
- (B) 1.5 in
- (C) 1.75 in
- (D) 2 in

46. A typical stress strain curve for mild steel is shown below. Which of the following is the correct descriptions of the points shown in the curve



- (A) A - Yield Point, C - Proportionality limit, D - ultimate strength, E - Fracture point
- (B) A - Proportionality limit, B - Elastic limit, C - Yield point, E - Fracture point
- (C) A - Elastic limit, C - Lower Yield point, D - Higher yield point, E - Ultimate strength
- (D) A - Proportionality limit, C - Elastic limit, D - Yield point, E - Fracture point

47. For a doubly symmetric member, the moment diagram between two braced points is shown below. What is the lateral torsional buckling modification factor C_b ?



- (A) 1
 (B) 1.67
 (C) 1.83
 (D) 2
48. Which of the following properties of structural timber is most significantly impacted due to the presence of knots?
- (A) Maximum density of the member,
 (B) Allowable compression parallel to the grain,
 (C) Allowable tension perpendicular to the grain,
 (D) Modulus of elasticity and stiffness

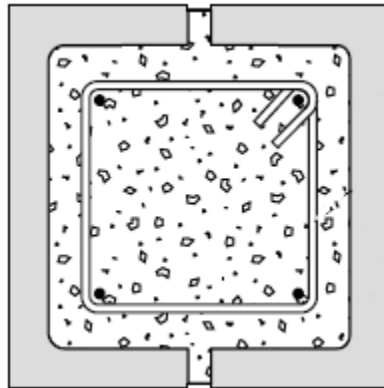
49. A 4X8 Alaska Yellow cedar No. 2 is to be used as a flexural member. What is the ASD adjusted value of bending for wind load conditions? Assume 110°F temperature and moisture content of 19 %. Take $C_i = 1$, $C_L = 0.9$.

- (A) 800 psi
- (B) 935 psi
- (C) 1127 psi
- (D) 1258 psi

50. An unreinforced, fully grouted masonry wall spans 12 ft vertically from the slab on grade to the roof level. The load bearing wall is laterally supported by cross walls at 10 ft center to center. What is the minimum thickness of the wall to be considered for design?

- (A) 4 in
- (B) 6 in
- (C) 8 in
- (D) 10 in

51. What is the minimum longitudinal reinforcement permitted in the masonry column shown below? The outer dimensions of the column are 9.625 in X 9.625 in.

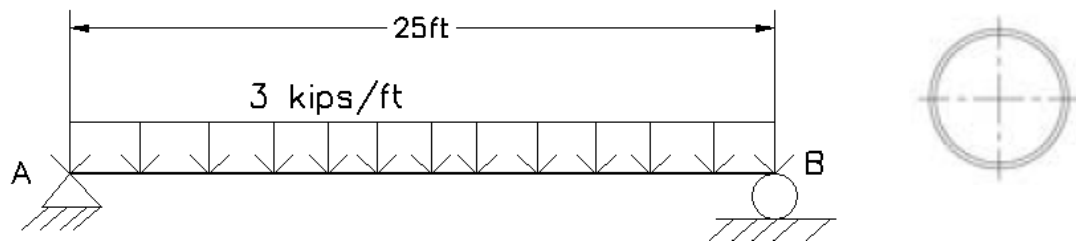


- (A) Two #3
- (B) Two #4
- (C) Three #3
- (D) Four #3

52. During a standard penetration test, the potential energy of the hammer before the drop was measured as 467 J and the energy transferred to the drill rod was 375 J. If the number of blows measured during the test was 8, what is the correct N_{60} value of blows?

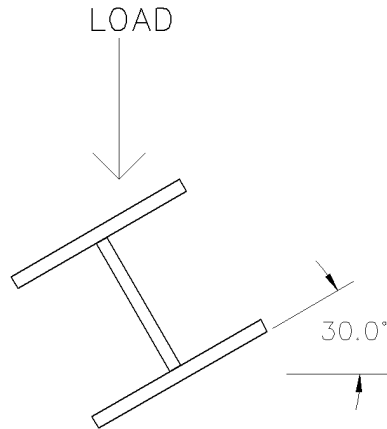
- (A) 6
- (B) 9
- (C) 10
- (D) 11

53. A simply supported steel beam has a span of 25 ft and is braced at the midpoint. The beam is subjected to a uniformly distributed live load of 3 kip/ft. What is the lightest ASTM A500 Gr. C, 46 round HSS section that will satisfy the flexural strength requirements?



- (A) HSS 16.000 X 0.375
- (B) HSS 16.000 X 0.438
- (C) HSS 18.000 X 0.375
- (D) HSS 18.000 X 0.5

54. A simply supported roof purlin has a span of 12 ft. The uniformly distributed load acting on the purlin is 1 kip/ft. If the allowable flexural strengths for bending in strong axis and weak axis are 30 kip*ft and 10 kip*ft respectively, what is the total flexural utilization of the section?



- (A) 50%
 (B) 75%
 (C) 90%
 (D) 150%
55. A built up section is to be used as a horizontal beam. The moment of inertia of the section about the stronger axis is 786 in^4 and about the weaker axis is 311.7 in^4 . The cross sectional area is 68 in^2 . What is the maximum unbraced length permitted in the beam section to prevent lateral torsional buckling?

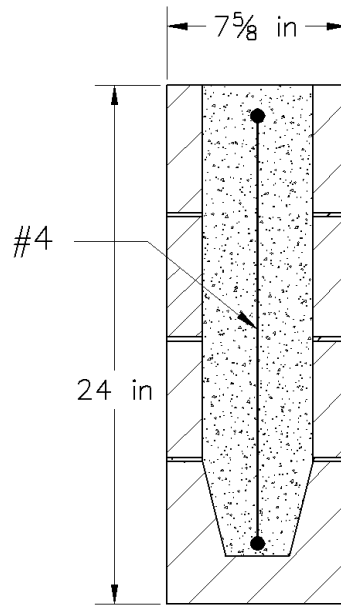
Material properties:

Yield Strength = 50 ksi

Modulus of Elasticity = 29000 ksi

- (A) 5 ft
 (B) 7.5 ft
 (C) 9 ft
 (D) 11.5 ft

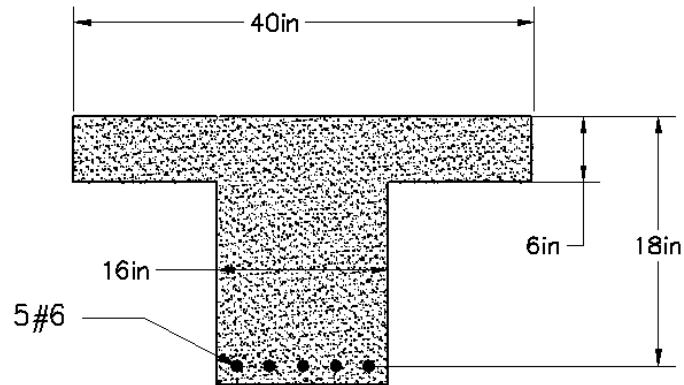
56. For the masonry beam section shown below, shear stress resisted by masonry, $F_{vm} = 43.6$ psi and shear stress F_{vs} resisted by shear reinforcement = 27.5 psi . If a single legged #4 bar is used as shear reinforcement, what is the maximum spacing of the #4 bar permissible? Effective depth for tension bar, $d = 20$ in.



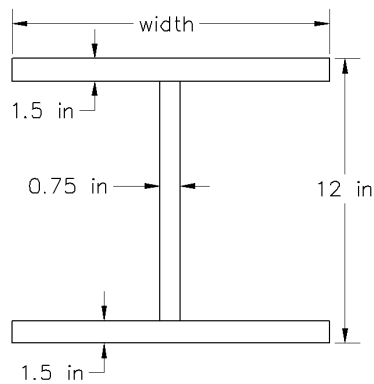
- (A) 6 in
(B) 9 in
(C) 15 in
(D) 24 in
57. A DFL select structural grade, simply supported beam has a span of 20 ft and is laterally unbraced. Determine the critical buckling design value F_{bE} for the beam if the beam slenderness $R_b = 8.3$. Assume wet conditions and normal temperature.
 $C_i = 1$ and buckling stiffness factor, $C_T = 1$

- (A) 8.1 ksi
(B) 9.7 ksi
(C) 10.8 ksi
(D) 12.5 ksi

58. What is the nominal moment capacity of the reinforced concrete T beam section shown below? The compressive strength of concrete is 4000 psi and the rebars are Grade 60.



- (A) 150 kip*ft
 (B) 170 kip*ft
 (C) 190 kip*ft
 (D) 230 kip*ft
59. For the built up column shown below, subjected to axial compression only, what is the maximum flange width permissible, so that the flanges are non slender?
 $F_y = 50$ ksi, $E = 29000$ ksi



- (A) 20 in
 (B) 28 in
 (C) 35 in
 (D) 40 in

60. A W21X111 Grade 60 column section spans 15 ft vertically between two floors. The column is unbraced in both directions. What is the design compressive strength for the column to be used in LRFD design?

$E = 29000 \text{ ksi}$

Effective length factor = 1

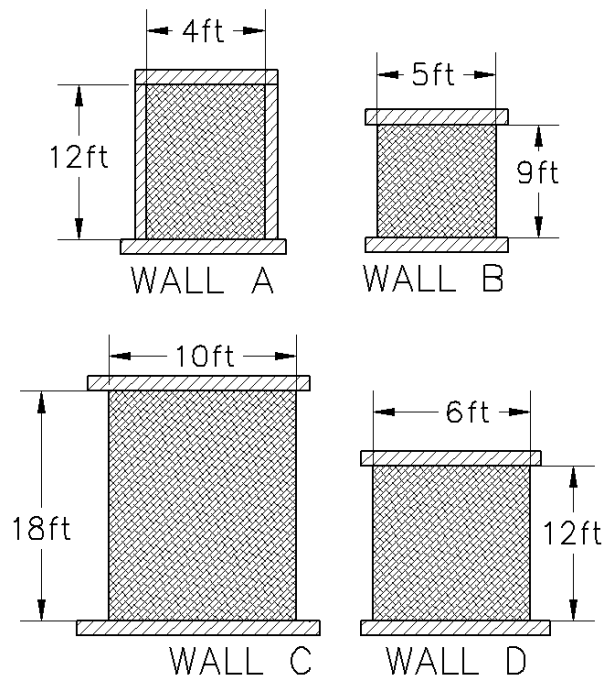
- (A) 1000 kips
- (B) 1050 kips
- (C) 1100 kips
- (D) 1250 kips

61. What is the allowable compressive force on a 23.625in X 23.625in masonry column section with specified compressive strength $f'_m = 2000 \text{ psi}$? The section has eight #8 Gr. 60 vertical rebars and is fully grouted. The effective height of the column is 20 ft. Neglect accidental eccentricity.

- (A) 325 kips
- (B) 385 kips
- (C) 457 kips
- (D) 650 kips

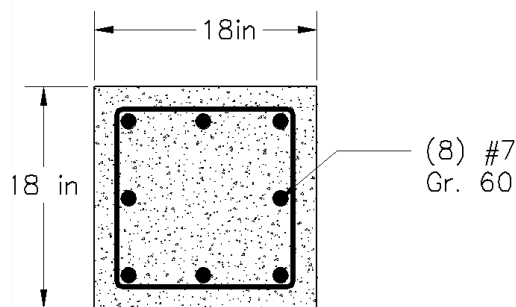
62. Which of the following WSP shear walls does not comply with dimensional requirements of AWC- SDPWS provisions for wood framed shear wall?

- (A) Wall A
- (B) Wall B
- (C) Wall C
- (D) Wall D



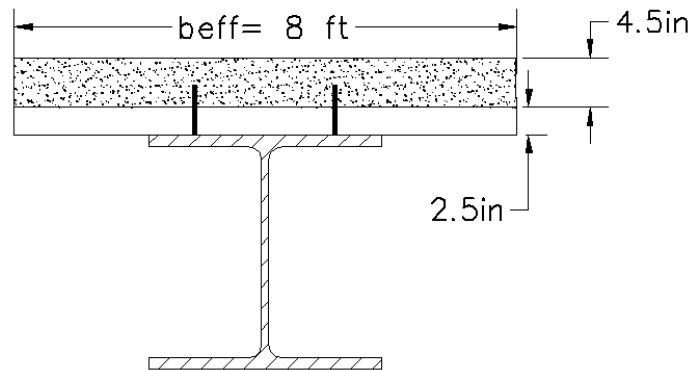
63. The column section shown below is subjected to a factored moment of 100 kip*ft. What is the design axial load capacity of the column?

$f'_c = 5 \text{ ksi}$, $f_y = 60 \text{ ksi}$, Area of #7 bar = 0.6 in^2 , $\Phi = 0.65$

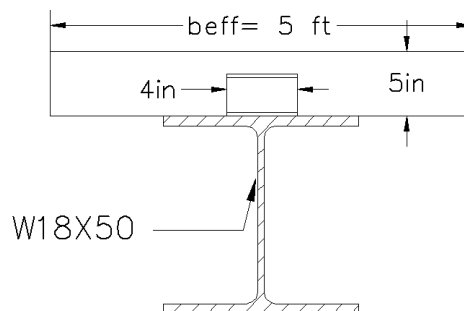


- (A) 855 kips
(B) 1125 kips
(C) 1315 kips
(D) 1630 kips
64. A thirty story building with a steel braced frame system is subjected to wind-induced vibrations. The building's natural frequency is close to the vortex shedding frequency of the wind. Which of the following is the most effective strategy to prevent structural damage due to large amplitude vibrations caused by resonance?
- (A) Increase the overall mass of the structure to counter wind loads and increase stability,
(B) Increase the height of the building that leads to altered natural frequency of the building,
(C) Install a tuned mass damper,
(D) Reduce the stiffness of the braces to alter the natural frequency of the building.

65. A composite steel beam with W18X50 steel section has the steel decking ribs perpendicular to the span of the beam as shown below. The modular ratio of steel to concrete is 8. What is the location of the elastic neutral axis (strong axis) of the composite member from the top of the composite section?



- (A) 5.19 in
(B) 6.23 in
(C) 11.32 in
(D) 16.11 in
66. A composite steel beam has four inch long channels, welded throughout the span to transfer horizontal shear. The beam is subjected to a factored dead load moment of 240 kip*ft and a factored live load moment of 400 kip*ft. The distance between the top of the steel beam and the concrete compression flange force is 4 in. Which of the following is most nearly the total nominal shear carried by the channels due to the applied moments? The capacity of the steel beam and concrete slab is sufficient.



- (A) 325 kips
(B) 430 kips
(C) 515 kips
(D) 760 kips

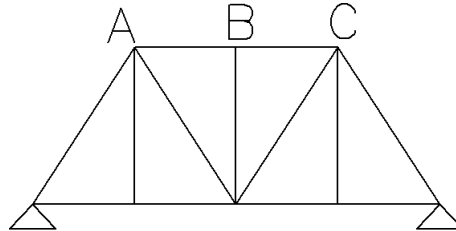
67. The truss shown below is subjected to external loads. After analysis, the internal forces in member BC for two service loading conditions are as follows:

Case A) Dead load = 5 kip tension

Live load =15 kip tension

Case B) Dead load = 5 kip tension

Live load = 18 kip compression



What is the most economical double angle section that satisfies the design requirement?

Span of the truss = 20ft and unbraced length of member BC = 8ft.

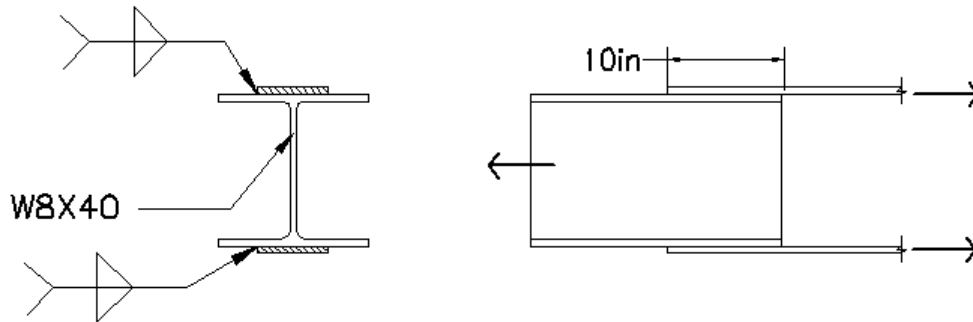
Gr. 36 Steel

- (A) 2 L 2X2X1/8
(B) 2 L 2X2X1/4
(C) 2 L 2X2X5/16
(D) 2 L 2X2X3/8

- 68.** A 0.5 inch thick plate is connected to the face of a column using 1/4 inch thick fillet welds. The total length of the end loaded fillet welds is 70 inches. What is the effective length of the fillet weld?

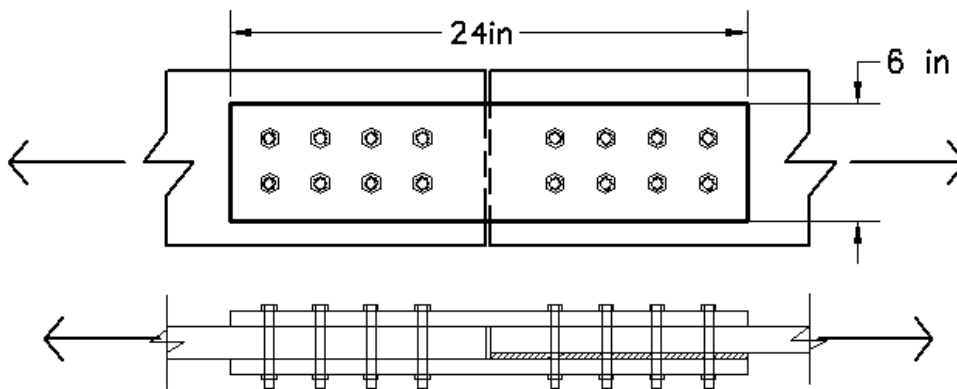
- (A) 37 in
(B) 42 in
(C) 45 in
(D) 58 in

69. Two plates are welded to the top and bottom of a W8X40 and tension load is applied as shown below. What is the shear lag factor “U” to calculate the net effective area of the section per AISC specifications?



- (A) 0.8
(B) 0.87
(C) 0.93
(D) 1

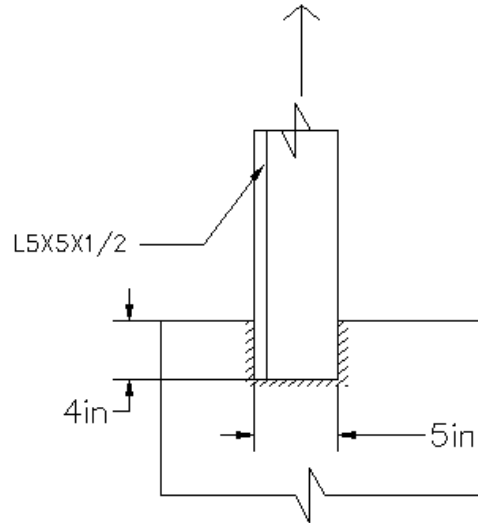
70. The plan and elevation of a tension splice is shown below. The tension load in the connection is 100 kips (factored). What is the required minimum thickness of the splice plate? $F_y = 36$ ksi and $F_u = 58$ ksi for the plate. Nominal hole dimension = $13/16$ in.



- (A) $1/4$ in
(B) $3/8$ in
(C) $1/2$ in
(D) $5/8$ in

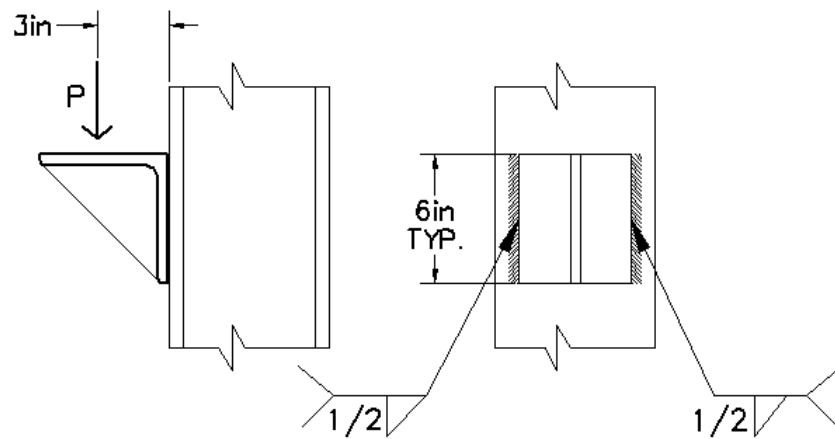
71. An angle section is connected to a plate by welds on three sides and subjected to a tensile loading in the direction shown below. The welds and plate do not govern failure of the connection. What is the design strength of the angle section? $F_y = 36$ ksi and $F_u = 58$ ksi for angle and plates.

- (A) 134.4 kips
(B) 155.2 kips
(C) 160.1 kips
(D) 175.5 kips



72. A bracket is welded on the face of the column as shown below. Using elastic method, what is the maximum allowable load that 'P' that the bracket can carry, assuming that the welds govern the failure of connection.

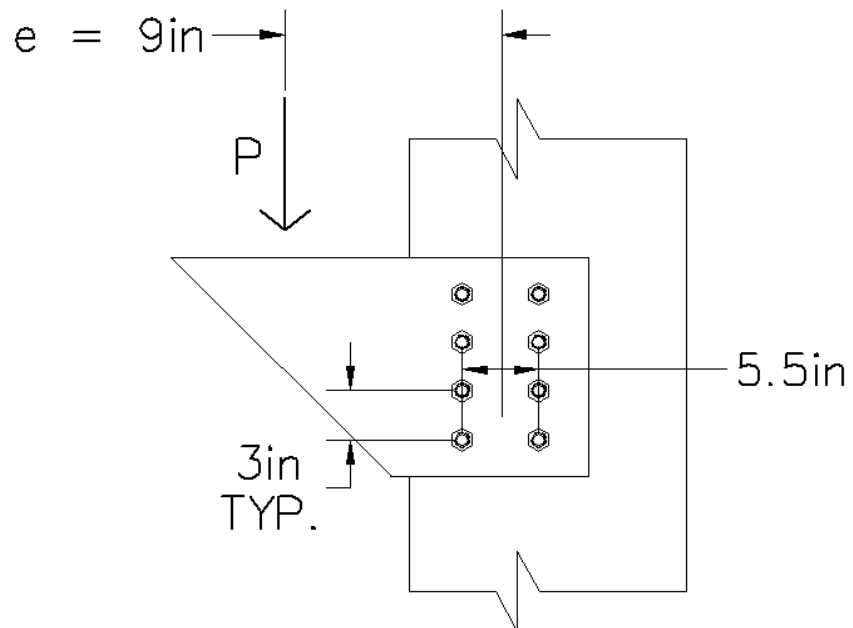
$F_{exx} = 70$ ksi



- (A) 17.5 kips
(B) 28.2 kips
(C) 32.2 kips
(D) 41.2 kips

73. A plate is bolted to a column with eight A490 Slip critical (Class A surface) bolts. The total load acting in the plate ' P ' = 20 kips, unfactored. What is the minimum diameter of the bolts required for standard size holes assuming bolt failure governs. Use instantaneous center of rotation method.

- (A) 1/2 in
(B) 5/8 in
(C) 3/4 in
(D) 1 in



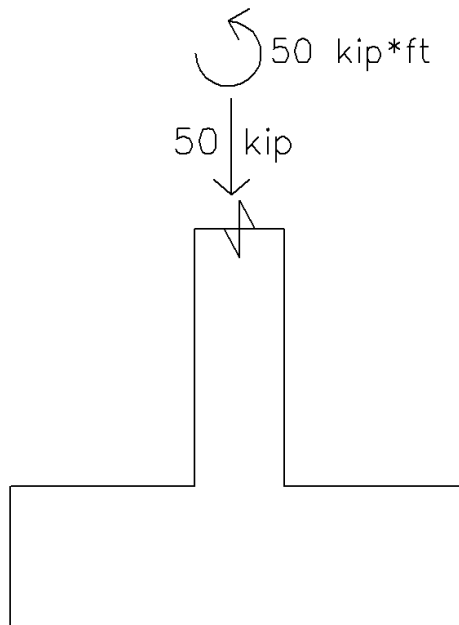
74. In a slip critical connection in single shear, a 3/4 in. diameter ASTM A325 bolt with standard hole is subjected to both shear and tension loads. The bolts are torqued per AISC requirements for slip critical connections. The tension load in the bolt is 5 kips. What is the allowable shear strength of the bolt due to shear tension interaction? Assume only one filler between the connected parts and class A surface.

- (A) 2.85 kips
(B) 3.61 kips
(C) 4.83 kips
(D) 6.33 kips

75. A 12 gauge steel plate is fastened to a 4X8 DFL member using 16d common nails. If the total lateral load to be transferred in the connection is 2800 lbs, what is the minimum number of nails required to secure the connection? Assume all adjustment factors are equal to 1.

- (A) 10
- (B) 15
- (C) 20
- (D) 25

76. A 5ft X 5ft square RCC footing bears on dense sandy soil. The concentric axial load acting on the footing is 50 kips including the footing weight. The superstructure column also transfers a moment of 50 kip*ft on the footing. What is the maximum bearing pressure acting under the footing?



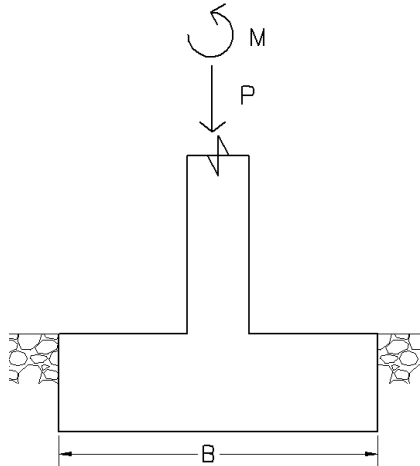
- (A) 2.3 ksf
- (B) 3.55 ksf
- (C) 4.44 ksf
- (D) 5.12 ksf

77. A square concrete footing is subjected axial loads and moments (including footing weight) from the column as follows:

Dead load = 100 kips Dead load moment = 50 kip*ft

Live load = 150 kips, Live load moment = 80 kip*ft

What should the minimum dimension 'B' of the footing be such that the entire footing area is effective in bearing the loads?



- (A) 2.75 ft
- (B) 3 ft
- (C) 3.25 ft
- (D) 4 ft

78. A 7ftX7ft square RCC footing is bearing on dense sand and has a depth of 18 inches. Per ACI 318-14 which of the following rebar layout is the most economical option that satisfies the minimum flexural rebar requirement for Gr 60 rebar?

$f'_c = 4 \text{ ksi}$ $\lambda = 1$

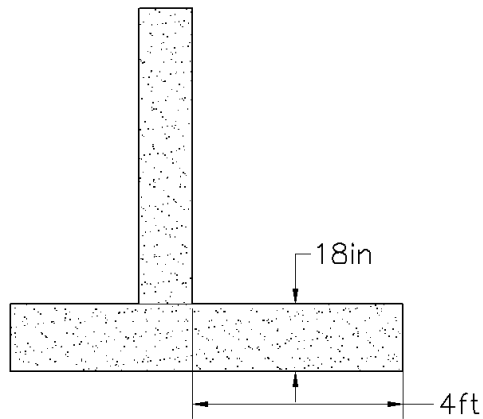
- (A) #4 @ 12 in. c/c
- (B) #5 @ 12 in. c/c
- (C) #6 @ 12 in. c/c
- (D) #7 @ 12 in. c/c

79. A pile is driven using a single acting air hammer weighing 700 lbs. The freefall height of the hammer is 12 ft. Every three blows from the hammer drives the pile 1 inch into the ground. What is most nearly the allowable load for the pile?

- (A) 15 ton
- (B) 19 ton
- (C) 23 ton
- (D) 30 ton

80. The heel of the retaining wall is subjected to 13583 lb*ft/ft factored flexural demand. What is the required area of tension reinforcement for the heel? Effective cover of the tension bars = 3.5in. $f'_c = 4.5$ ksi , $F_y = 60$ ksi

- (A) 0.201 in²/ft
- (B) 0.583 in²/ft
- (C) 0.671 in²/ft
- (D) 0.852 in²/ft



Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (B) 61 kip

Explanation: {Ref NCEES manual page 24 for area of common shapes}

Area of trapezoidal cross section = $0.5 * \text{height} * (\text{sum of opposite parallel sides})$

$$= 0.5 * 10\text{ft} * (2\text{ft} + 4\text{ft})$$

$$= 30\text{ft}^2$$

Length of abutment = 12ft

Volume of abutment = Area*Length

$$= 12\text{ft} * 30\text{ft}^2$$

$$= 360 \text{ ft}^3$$

Density of stone masonry = 0.17 kcf

{Ref AASHTO LRFD Table 3.5.1-1 Unit weights}

Weight of abutment = density *volume

$$= 0.17 \text{ kcf} * 360 \text{ ft}^3$$

$$= 61.2 \text{ kip} \approx \mathbf{61 \text{ kip}}$$

2. Problem

Correct Answer: (B) 1282 lb

Explanation: {Ref AISC Steel manual page 1-97 and page 1-99 Table 1-12 for weight of HSS sections}

Weight of HSS 6X6X1/2 = 35.24 lb/ft

Weight of HSS 2X2X1/4 = 5.41 lb/ft

Total length of vertical and horizontal HSS 6X6X1/2 members = $(10+6+6+4+4) \text{ ft} = 30\text{ft}$

Length of individual diagonal members = $\sqrt{3^2 + 4^2} \text{ ft} = 5\text{ft}$

Total length of diagonal members = $(5+5+5+5)\text{ft} = 20\text{ft}$

Total weight of assembly = Weight of 6X6X1/2 members + Weight of 2X2X1/4 members

$$= 30\text{ft} * 35.24\text{lb/ft} + 20\text{ft} * 5.41 \text{ lb/ft}$$

$$= 1165.4 \text{ lb}$$

Capacity of hoisting mechanism required = Total weight + 10% Total weight for rigging

$$= 1.1 * 1165.4 \text{ lb} = 1281.91 \text{ lb} \approx \mathbf{1282 \text{ lb}}$$

3. Problem

Correct Answer: (C) **12.2 kips**

Explanation:

Total live load on ground floor column

$$= (0.5 \text{ kips/ft} + 1.2 \text{ kips/ft}) * 15 \text{ ft} / 2$$

$$= 12.75 \text{ kips}$$

$$\text{Tributary area} = 112.5 \text{ ft}^2$$

$K_{LL} = 4$ Live load element factor for exterior column From Table 4.7-1

$$K_{LL} * A_T = 4 * 112.5 = 450 \text{ ft}^2$$

$$L = L_o \left(0.25 + \frac{15}{\sqrt{K_{LL} * A_T}} \right)$$

$$L = 12.75 \text{ kips} \left(0.25 + \frac{15}{\sqrt{4 * 112.5}} \right) = \mathbf{12.2 \text{ kips}}$$

4. Problem

Correct Answer: (C) Sloped roofs have reduced accessibility and are less likely to be occupied

Explanation:

Strength of roofing members depend on materials and section properties, not inclination,

Snow loads and seismic loads have nothing to do with reduction factors of live loads.

Hence the correct answer is because of decreased access in roofs with higher slopes, loads are permitted to be reduced.

5. Problem

Correct Answer: (A) 5 kips/ft

Explanation:

Unit weight of concrete $\gamma = 150$ pcf, Lateral pressure exerted by concrete $= \gamma(10\text{ft})$

Total lateral load exerted by 10ft high fresh concrete

$$L = 0.5 * \text{Lateral Pressure} * \text{height} = 0.5 * \gamma(10\text{ft})^2$$

$$L = 0.5 * 150 \text{ lb/ft}^3 * 100 \text{ ft}^2 \quad \text{This gives, } L = 7500 \text{ lb/ft}$$

Following triangular distribution, the point of application of pressure is at $10/3$ ft from the base = 3.33ft

Taking moments about the base to be zero, the lateral force P at the point of bracing is

$$P * 5\text{ft} = 7500 \text{ lb/ft} * 3.33\text{ft}$$

$$P = 4995 \text{ lb/ft} \approx \mathbf{5 \text{ kip/ft}}$$

6. Problem

Correct Answer: (C) Exposure C

Explanation:

Per ASCE 7-16 page 266, section 26.7, the surface with obstructions qualifies for Exposure category B, However, it only extends upto 2500 ft, beyond which the surface is Exposure C.

For buildings or other structures with a mean roof height greater than 30 ft (9.1 m),

Exposure B shall apply where Surface Roughness B prevails in the upwind direction for a distance greater than 2,600 ft (792 m) or 20 times the height of the building or structure, whichever is greater. Hence the structure does not qualify for category B and the correct answer is category C.

7. Problem

Correct Answer: (B) 30 kips

Explanation:

ASCE 7-16 chapter 13 gives provisions for calculation of seismic demands on non structural component

Weight $W_p = 50\text{kips}$

Design short period spectral acceleration $S_{DS} = 0.5g$

Importance factor $I_p = 1$

From page 126 table 13.5-1

Response modification factor for unbraced parapet $R_p = 2.5$

Amplification factor for unbraced parapet $a_p = 2.5$

The seismic demand is given by page 123 Equation 13.3-1

$$F_p = \frac{0.4 * a_p * S_{DS} * W_p}{\frac{R_p}{I_p}} \left(1 + 2 \frac{z}{h} \right)$$

Since component is at the roof of the building, $z = h$

$$F_p = \frac{0.4 * 2.5 * 0.5 * 50 \text{ kips}}{\frac{2.5}{1}} \left(1 + 2 \frac{h}{h} \right) = 30 \text{ kips}$$

8. Problem

Correct Answer: (D) Design truck or Design tandem plus 0.64 klf of uniformly distributed lane load

Explanation:

Per AASHTO LRFD section 3.6.1.2, the total design vehicular live load is the combination of either the design truck and design lane load, or the design tandem and design lane load.

Per section 3.6.1.2.4 the design lane load is a load of 0.64 klf uniformly distributed in the longitudinal direction. Hence, option D is the correct answer

9. Problem

Correct Answer: (D)

Explanation:

From ASCE 7-16 page 54, section 7.5.1, all other loading conditions are valid for partial except loading condition D. The provisions don't mention applying half the uniform loading on all spans.

10. Problem

Correct Answer: (A) 24.6 lb/ft

Explanation:

From ASCE 7-16, page 5, for risk category III,

Ice importance factor $I_i = 1.15$

Given

Height of roof, $z = 100$ ft

Nominal ice thickness, $t = 1.5$ inches

Topography factor $K_{zt} = 1$

For L6X6X1/2 from figure 10.4-1, page 70

$$D_c = \sqrt{6^2 + 6^2} = 8.49 \text{ inches}$$

From page 70, equation 10.4-4, height factor

$$f_z = \left(\frac{z}{33}\right)^{0.1} = \left(\frac{100}{33}\right)^{0.1} = 1.117$$

Design ice thickness for freezing rain from page 71, equation 10.4-5

$$t_d = t * I_i * f_z * K_{zt}^{0.35} = 1.5 \text{ inch} * 1.15 * 1.117 * 1 = 1.93 \text{ inches}$$

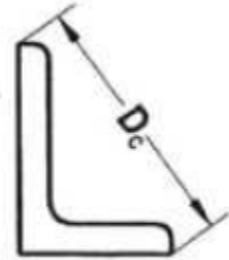
Area of ice forming in structural shapes, prismatic members is given by page 70, equation 10.4-1

$$A_i = \pi * t_d (D_c + t_d) = \pi * 1.93(8.49 + 1.93) = 63.18 \text{ sq. in}$$

From section 10.4.1

Density of ice = 56 pcf

Thus, load due to weight of ice = 56 pcf * (63.18/144) sq.ft = **24.6 lb/ft**



11. Problem

Correct Answer: (D)

Explanation:

Options A, B and C are correct per ASCE 7-16 page 16-18, section 4.6 and section 4.9.

Option D is incorrect because in addition to the recommended impact factors, for LRFD design, the loads need to include an additional 1.6 factor for live load and dead load combinations.

12. Problem

Correct Answer: (B) 1848 lb/ft

Explanation:

The coefficient to be used here is earth pressure at rest

Earth pressure coefficient at rest, $K_o = 0.5$

Density of soil, $\gamma = 110$ pcf

From table 3.11.6.4-1, page 3-142, the equivalent height of soil for vehicular loadings are

For 10 ft abutment = 3ft For 20ft abutment = 2ft (Abutments perpendicular to traffic)

Interpolating for 12ft, $h_{eq} = 2.8$ ft

Then, from equation 3.11.6.4-1, surcharge loading pressure intensity

$$= K_o \cdot \gamma \cdot h_{eq} = 0.5 \cdot 110 \text{ pcf} \cdot 2.8 \text{ ft} = 154 \text{ psf}$$

Total load = area of rectangular pressure diagram of height 12ft

$$= 12 \text{ ft} \cdot 154 \text{ psf} = \mathbf{1848 \text{ lb/ft}}$$

13. Problem

Correct Answer: (B) 2666.67 lbs

Explanation:

Reinforced concrete floor is idealized as rigid diaphragm.

A rigid diaphragm distributes load proportional to stiffness

$$\text{Total load acting on diaphragm} = (500 \text{ lb/ft} \cdot 20 \text{ ft}) = 10000 \text{ lbs}$$

Load shared by central wall

$$= 10000 \cdot 4 \text{ ft} / (4 \text{ ft} + 3 \text{ ft} + 3 \text{ ft} + 2.5 \text{ ft} + 2.5 \text{ ft})$$

$$= \mathbf{2666.67 \text{ lbs}}$$

14. Problem

Correct Answer: (C) 55 kips

Explanation:

D = 25 kips

L = 5 kips

$L_r = 10$ kips

S = 8 kips (does not govern over roof live load)

W = 15 kips

Roof live load is greater than live load from equipment, so combination 2 won't govern.

Check governing strength design combinations.

From ASCE 7-16, section 2.3, page 7

Combination 3

$$1.2D + 1.6L_r + 0.5W = 1.2*25 + 1.6*10 + 0.5*15 = 53.5 \text{ kips}$$

Combination 4

$$1.2D + 1W + L + 0.5L_r = 1.2*25 + 1*15 + 5 + 0.5*10 = 55 \text{ kips}$$

Hence governing factored load is **55 kips**

15. Problem

Correct Answer: (C)

Explanation:

The bending moment at the hinge located at the third span must always be zero.

Hence this rules out options A and options B.

Option D does not show any changes in the slope of the bending moment diagram in the first and third spans when there is a concentrated load. Hence Option D is not correct.

Thus the correct answer is Option C.

16. Problem

Correct Answer: (A)

Explanation:

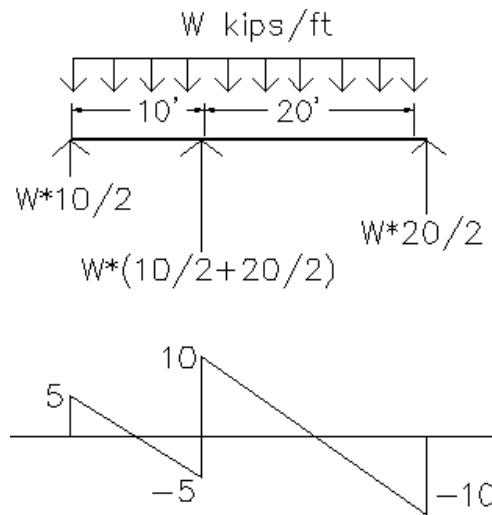
Since the diaphragm is flexible, the shear wall load distribution is per the tributary area.

If 'W' is the diaphragm load W kip/ft,

The left wall shear is $10\text{ft}/2 * W = 5W$

The center wall shear is $(10\text{ft}/2 + 20\text{ft}/2)W = 15W$

The right wall shear is $20\text{ft}/2 * W = 10W$



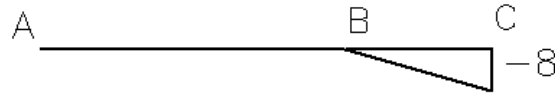
This can be thought of as a two span beam with 3 reactions as shown above. The correct shear force diagram is Option A for this beam.

17. Problem

Correct Answer: (D)

Explanation:

The moment at support B will be zero when the load is travelling between the two supports i.e. point A to point B. When the load is travelling along the overhanging portion, the maximum moment occurs when the load is at the edge of the overhang and zero when the load is directly on top of B. Thus it will have a linear distribution between B and C as shown below :



The correct answer is Option D.

18. Problem

Correct Answer: (B) 860 lbs

Explanation:

Let the point of attachment of the jib to the mast be 'A', the point of attachment of tie be point 'B' and the point of suspended load be point 'C'.

Let T be the tension in the Tie

Angle of inclination of tie to the horizontal

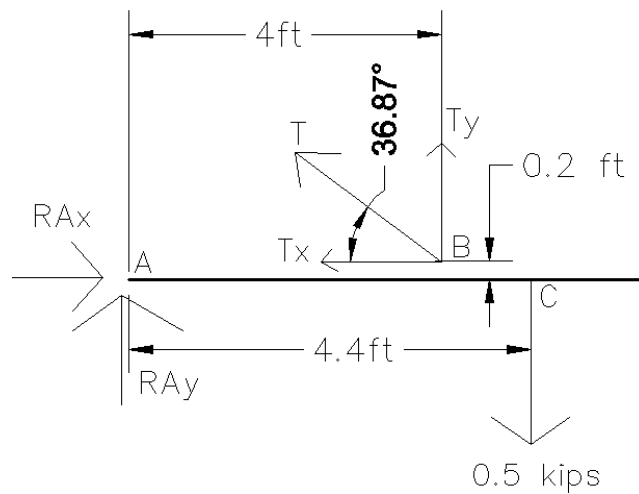
Horizontal distance between tie and upper support = 4ft

Vertical height between tie and upper support = 3.2ft - 0.2ft = 3ft

$$\Theta = \tan^{-1}(3\text{ft} / 4\text{ft}) = 36.87^\circ$$

Horizontal component of tension, $T_x = T \cos \Theta$

Vertical component of tension, $T_y = T \sin \Theta$



Taking sum of moments about reaction point 'A'

$$T_x * 0.2\text{ft} + T_y * 4\text{ft} = 0.5 \text{ kips} * 4.4\text{ft}$$

$$T * \cos(36.87^\circ) * 0.2\text{ft} + T * \sin(36.87^\circ) * 4\text{ft} = 0.5\text{kip} * 4.4\text{ft}$$

$$T = 0.859 \text{ kips} \approx \mathbf{860 \text{ lbs}}$$

Thus the correct answer is option B.

19. Problem

Correct Answer: (B) 450 kips

Explanation:

Here , total number of piles , $n = 5$

Total axial load = 2000 kips

Moments about X-X axis, $M_{xx} = 500 \text{ kip}\cdot\text{ft}$

Moments about Y-Y axis, $M_{yy} = 300 \text{ kip}\cdot\text{ft}$

The pile layout is symmetric and the centroid of pile group is at the center of middle pile

Let x and y be the distances of each pile from the centroid of the group.

$$\sum x^2 = (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (0) = 64 \text{ ft}^2$$

$$\sum y^2 = (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (4\text{ft})^2 + (0) = 64 \text{ ft}^2$$

Axial load distribution in each pile

$$= 2000 \text{ kips}/n$$

$$= 2000 \text{ kips}/5 = 400 \text{ kips}$$

Additional axial load due to M_{xx}

$$= M_{xx} * (x/\sum x^2) = 500 \text{ kip}\cdot\text{ft} * 4\text{ft}/64 \text{ ft}^2 = 31.25 \text{ kips}$$

Additional axial load due to M_{yy}

$$= M_{yy} * (y/\sum y^2) = 300 \text{ kip}\cdot\text{ft} * 4\text{ft}/64 \text{ ft}^2 = 18.75 \text{ kips}$$

$$\text{Max total axial load} = 400 \text{ kips} + 31.25 \text{ kips} + 18.75 \text{ kips} = \mathbf{450 \text{ kips}}$$

Correct answer is option B.

20. Problem

Correct Answer: (A)

Explanation:

The given options of trusses B, C and D have mixed forces i.e. tension and compression in the diagonals.

Option A is a typical Howe truss in which the diagonals are only subjected to compressive forces. Hence Option A is the correct answer.

21. Problem

Correct Answer: (C) 92 kips

Explanation:

Per ACI 318-14 section 18.6.5 page 279, for special moment resisting frames, the beams shall be able to develop the shear due to factored gravity loads and maximum probable moment acting at the joint faces

Factored Dead load = $1.2 \times 2 \text{ kip/ft} = 2.4 \text{ kip/ft}$

Factored Live load = $1 \times 1 \text{ kip/ft} = 1 \text{ kip/ft}$

Total $w_u = 2.4 + 1 = 3.4 \text{ kip/ft}$

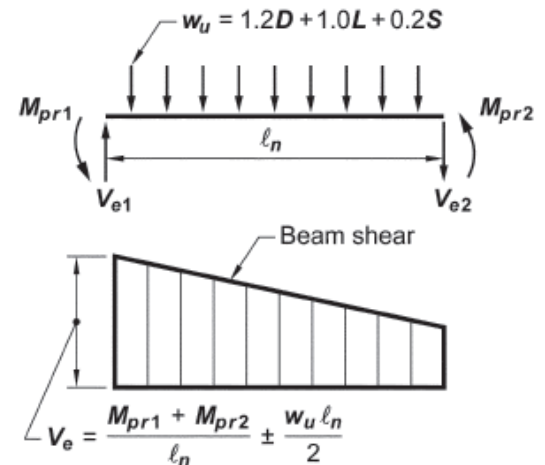
$M_{pr1} = 500 \text{ kip}\cdot\text{ft}$ $M_{pr2} = 600 \text{ kip}\cdot\text{ft}$

Clear span, $\ell_n = 18 \text{ ft}$

Maximum shear

$$\begin{aligned} &= (M_{pr1} + M_{pr2})/\ell_n + w_u \ell_n / 2 \\ &= (500 \text{ kip}\cdot\text{ft} + 600 \text{ kip}\cdot\text{ft}) / 18 \text{ ft} + (3.4 \text{ kip/ft} \times 18 \text{ ft}) / 2 \\ &= 91.7 \text{ kip} \approx \mathbf{92 \text{ kips}} \end{aligned}$$

Hence, the correct answer is option C.



22. Problem

Correct Answer: (C)

Explanation:

Ref to NCEES PE handbook page 32 for shear flow distribution.

For a rectangular section, shear stress does not vary linearly between the top and bottom.

Shear stress is zero at the top and bottom and maximum at the neutral axis.



Hence the correct answer is Option C

23. Problem

Correct Answer: (D) 16.64 kips

Explanation:

Calculate the maximum reaction at pinned end per unit spacing of floor beam

(Taking moments about roller = 0)

$$R_a \cdot 20\text{ft} = 0.8 \text{ kips} \cdot 20 \text{ ft} + 0.16 \text{ kip/ft} \cdot 30\text{ft} \cdot 5\text{ft} - 0.6 \text{ kip} \cdot \text{ft} - 0.1 \text{ kip} \cdot 10\text{ft}$$

$$R_a = 1.92 \text{ kips}$$

Reaction at roller end (Sum of vertical forces = 0)

$$R_b = 0.8 + 0.16 \cdot 30 + 0.5 + 0.1 - R_a = 4.28 \text{ kips}$$

Thus maximum shear occurs at R_b and the value is

$$V_b = 0.1 + 0.16 \cdot 10 + 0.5 - R_b = -2.08 \text{ kips (calculated per unit spacing)}$$

For floor beams spaced at 8 ft c/c, total shear = $2.08 \cdot 8 = \mathbf{16.64 \text{ kips}}$

Thus, the correct answer is option D.

24. Problem

Correct Answer: (C) 90 kip *ft

Explanation:

Bending moment at the base of the column due to lateral load

$$M = 5 \text{ kips} \cdot 15 \text{ ft}$$

$$M = 75 \text{ kip} \cdot \text{ft}$$

Lateral stiffness of the column = 2.5 kips/in

Thus displacement at the top of the column due to lateral load of 5 kips

$$d = 5 \text{ kips} / (2.5 \text{ kips/in})$$

$$d = 2 \text{ in}$$

This deflection causes additional second order loading effects due to the axial load and increases the bending moment at the base.

Additional moment

= Axial load * deflection (i.e. eccentricity for the axial load)

$$= 85 \text{ kips} \cdot 2 \text{ in} \cdot (1/12 \text{ ft/in})$$

$$= 14.17 \text{ kip} \cdot \text{ft}$$

$$\text{Total moment} = 75 \text{ kip} \cdot \text{ft} + 14.17 \text{ kip} \cdot \text{ft} = 89.17 \text{ kip} \cdot \text{ft} \approx \mathbf{90 \text{ kip} \cdot \text{ft}}$$

Thus the correct answer is option C.

25. Problem

Correct Answer: (D) 2.125 in

Explanation:

The plastic neutral axis divides the areas of tension and compression into two equal parts.

Let the plastic neutral axis be a distance x from the top.

Total area of the section

$$= 4\text{in} \cdot 0.5\text{in} + 2\text{in} \cdot 0.5\text{in}$$

$$= 3\text{ in}^2$$

If area is divided into half,

Area of upper half = $A/2$

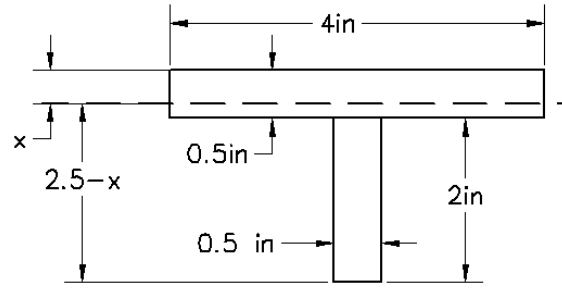
$$4\text{in} \cdot x = 3\text{ in}^2 / 2$$

$$x = 0.375\text{ in}$$

Location of plastic neutral axis from base = $2.5\text{ in} - x$

$$= 2.5\text{ in} - 0.375\text{ in} = \mathbf{2.125\text{ in}}$$

Thus correct answer is option D



26. Problem

Correct Answer: (B) 2.3 kip*ft

Explanation:

Ref NCEES PE handbook page 232 for distribution factors and page 241-244 for FEMs.

E is constant and not considered in the calculation since it cancels out in numerator and denominator for calculation of distribution factor.

Calculate joint stiffness and distribution factors for joint B (units and E not shown for calculation simplification).

Ref to NCEES PE handbook page 232 section 4.1.3, rotational stiffness “k” for far end fixed condition is given by $k = 4EI/L$

If the far end is hinged instead of fixed, $k = 3EI/L$

$$K_{bc} = 3 \cdot 800 / (12 \cdot 12) = 16.67$$

$$K_{ba} = 4 \cdot 550 / (16 \cdot 12) = 11.46$$

$$K = 16.67 + 11.46 = 28.13$$

$$DF_{bc} = 16.67 / 28.13 = 0.593$$

$$DF_{ba} = 11.46 / 28.13 = 0.407$$

	Joint A	Joint B		Joint C
	AB	BA	BC	CB
DF	0	0.407	0.593	0
FEM	$-(4 \cdot 8 \cdot 8^2) / 16^2$ = -8	$(4 \cdot 8 \cdot 8^2) / 16^2$ = 8	$-2 \cdot 12^2 / 8$ = -36	
Balancing moment		28		
Distribution		11.39	16.6	
Carryover	5.695			
Resultant	-2.305	19.39	-19.4	0

Hence, correct answer is option B, 2.3 kip*ft.

27. Problem

Correct Answer: (C)

Explanation:

When an object is in pure shear

Principal stresses, $\sigma_1 = \tau$ $\sigma_2 = -\tau$, that is $\sigma_2 = -\sigma_1$ and $\sigma_3 = 0$

Out of the three stress matrices, Option C resembles this stress state.

Thus, Option C is the correct answer.

28. Problem

Correct Answer: (D) 14.54 kip/in

Explanation:

Maximum bending moment acting in the weld

$$M = 20 \text{ kip} \cdot 12 \text{ in}$$

$$M = 240 \text{ kip} \cdot \text{in}$$

The full penetration weld can be treated as a linear weld with height $h = 10 \text{ in}$ and unit width

$$\text{Moment of inertia, } I = (1) \cdot (10 \text{ in})^3 / 12 = 83.333 \text{ in}^3$$

$$\text{Section modulus, } S = 83.333 \text{ in}^3 / (10 \text{ in} / 2) = 16.67 \text{ in}^3$$

Maximum tension acting on the weld

$$T = M / S$$

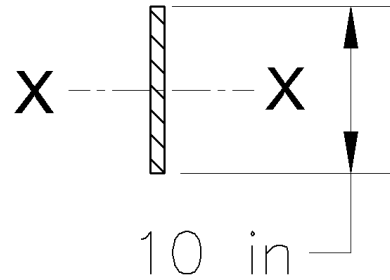
$$T = 240 \text{ kip} \cdot \text{in} / 16.67 \text{ in}^3 = 14.4 \text{ kip/in}$$

Maximum shear acting on the weld

$$H = 20 \text{ kip} / 10 \text{ in} = 2 \text{ kip/in}$$

Resultant stress

$$= \sqrt{(14.4 \text{ kip/in})^2 + (2 \text{ kip/in})^2} = \mathbf{14.54 \text{ kip/in}} \quad \text{Correct answer is option D}$$



29. Problem

Correct Answer: (D)

Explanation:

When a threaded rod is tightened, a torque is applied, however, it does not cause any bending in the rod. It causes pretension in the rod which is balanced by the clamping force via bearing under washers and nuts. Thus option D is the correct answer.

30. Problem

Correct Answer: (A) 0.16 in

Explanation:

Ref TMS 402 page C-52, section 5.2.1.4.1.

Deflection limits for beams and lintels supporting unreinforced masonry is $L/600$

$$L = 8\text{ft}$$

Allowable deflection

$$= L/600$$

$$= 8\text{ft} \times 12/600$$

$$= \mathbf{0.16\text{ in}}$$

Correct answer is option A.

31. Problem

Correct Answer: (B) 0.017 in

Explanation:

Ref NCEES PE handbook page 233 for truss deflection .

Forces on members due to 30 kips acting towards right on joint B and their lengths,

$$F_{ab} = -15\text{ kips} \quad L_{ab} = 120\text{ in}$$

$$F_{bd} = 25\text{ kips} \quad L_{bd} = 180\text{ in}$$

$$F_{ad} = 10\text{ kips} \quad L_{ad} = 100\text{ in}$$

$$F_{cd} = 0$$

$$F_{bc} = 0$$

Since the desired deflection is at joint B, unit load has to be applied at joint B.

Forces on members due to 1 kip acting towards right (can be directly scaled for this case)

$$f_{ab} = -15/30 = -0.5$$

$$f_{bd} = 25/30 = 0.833$$

$$f_{ad} = 10/30 = 0.333$$

$$f_{cd} = 0$$

$$f_{bc} = 0$$

Deflection at joint is given by

$$\Delta = \sum f \cdot (F \cdot L / AE)$$

Since $A=10 \text{ in}^2$ and $E = 29000 \text{ ksi}$ is constant for all members,

$$\sum f \cdot F \cdot L = (-0.5) \cdot (-15) \cdot 120 + 0.833 \cdot 25 \cdot 180 + 0.333 \cdot 10 \cdot 100$$

$$= 4981.5$$

$$\Delta = 4981.5 / (10 \cdot 29000)$$

$$\Delta = \mathbf{0.017 \text{ in}}$$

Correct answer is option B.

32. Problem

Correct Answer: (B) 0.03 in

Explanation:

Ref to NCEES PE handbook page 28 for axial deflection.

The three bars are parallel to each other and the stiffness of the system is the sum of stiffness of all three bars.

$$K_1 = AE/L$$

$$K_2 = 2AE/L$$

$$K_3 = AE/L$$

$$\text{Total stiffness} = k_1 + k_2 + k_3$$

$$= AE/L + 2AE/L + AE/L$$

$$= 4AE/L$$

$$\text{Total deflection} = \text{Load} / \text{stiffness}$$

$$= P / (4AE/L)$$

$$= PL / 4AE$$

$$= 200 \text{ kips} \cdot 180 \text{ in} / (4 \cdot 10 \text{ in}^2 \cdot 29000 \text{ ksi})$$

$$= \mathbf{0.031 \text{ in}} \quad \text{Correct answer is option B.}$$

33. Problem

Correct Answer: (B) 0.096

Explanation:

Ref to NCEES PE handbook page 31 for angle of twist and page 28 for shear modulus and page 25 for polar moment of inertia.

$$G = E/2(1+\nu) = 29000 \text{ ksi} / 2(1+0.3) = 11153.8 \text{ ksi}$$

Since the torque is applied at the center, torque is resisted equally by the two supports.

$$T = 500 \text{ kip}\cdot\text{ft} / 2 = 250 \text{ kip}\cdot\text{ft} = 3000 \text{ kip}\cdot\text{in}$$

$$\text{Polar moment of inertia, } J = \pi(8 \text{ in})^4 / 32 = 402.12 \text{ in}^4$$

$$\text{Angle of twist} = TL/GJ = (3000 \cdot 144) / (11153.8 \cdot 402.12) = \mathbf{0.096} \quad \text{Correct answer is Option B.}$$

34. Problem

Correct Answer: (D)

Explanation:

Static and dynamic analysis needs to be done in an analytical model without the failed column to ensure that all structural members sufficiently withstand the redistribution of loads. The bearing and uplift under the failed column does not need to be checked and is the correct answer.

35. Problem

Correct Answer: (C) 13/16 inch.

Explanation:

Ref to 'Specification for Structural Joints Using High-Strength Bolts-2014', page 16.2-23

Table 3.1 (part 16 of AISC steel manual 15th Ed.)

For 5/8 inch diameter bolts, oversized diameter of holes = **13/16 inch.**

Hence the correct answer is Option C.

36. Problem

Correct Answer: (A)

Explanation:

Ref to IBC 2018 page 420, Table 1705.3

Verifying rebar placement is under periodic inspection.

Adhesive anchors installed horizontally require continuous inspection.

Groove welds require continuous inspection.

The process of prestressing concrete members requires constant monitoring.

Hence the correct answer that requires intermittent i.e. periodic inspection is Option A.

37. Problem

Correct Answer: (B)

Explanation:

'Approved as noted' typically means that minor discrepancies found in the submittal package is acceptable/tolerable and the contractor can proceed with further construction work in the field. There is no requirement for resubmissions. Hence, the correct answer is Option B.

38. Problem

Correct Answer: (B)

Explanation:

Reshoring is done after the initial setting and initial removal of formworks. This ensures the concrete slab can continue curing and gain full strength and redistribute any loads acting on it to the floors below and the foundation. Hence the correct answer is option B.

39. Problem

Correct Answer: (C) 9ft

Explanation:

Difference between depths of
foundation between two structures

$$= 15 \text{ ft} - 6 \text{ ft}$$

$$= 9 \text{ ft}$$

For a 45 degree line,

Horizontal Distance

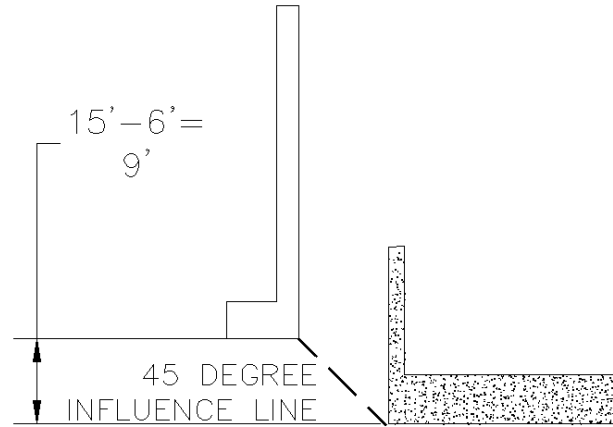
$$= 9 \text{ ft} * \tan (45 \text{ degree})$$

$$= 9 \text{ ft}$$

Thus, the minimum distance between
the two structures for existing

structure to be away from influence of new structure = 9ft

Correct answer is option C.



40. Problem

Correct Answer: (C)

Explanation:

Per IBC 2018, page 609, section 3307, "Protection of adjoining property",

All options are correct except Option C.

IBC 2018 Chapter 33 only says that excavation should be done in a way to not endanger life or property. Vacating adjacent buildings is optional but not mandatory. Hence, the correct answer is option C.

41. Problem

Correct Answer: (B) Good

Explanation:

Ref to NCEES PE handbook page 120, section 3.7.4.1. for rock quality designation

Total Length of core run = 6 ft = 72in

Length of sound core pieces >4 in

$$= 19+8+11+6+6+12$$

$$= 62\text{in}$$

RQD = sum of length of sound core pieces/ total core run length

$$= 62 \text{ in} / 72 \text{ in}$$

$$= 0.86$$

$$= 86\% \text{ (Good from table in page 120)}$$

Thus the correct answer is Option B, **Good**.

42. Problem

Correct Answer: (B) 99 days

Explanation:

Ref to NCEES PE handbook page 85 section 3.2.3 for time rate of settlement.

Coefficient of consolidation

$$C_v = 0.29 \text{ ft}^2 / \text{day}$$

$$\text{Depth of clay layer} = 155 \text{ ft} - 135 \text{ ft} = 20 \text{ ft}$$

Longest distance to drainage boundary

$$H_d = 20 \text{ ft} / 2 = 10 \text{ ft}$$

From table in page 86, for $U = 60\%$, Time factor

$$T_v = 0.287$$

From equation in page 85

$$T = T_v * (H_d)^2 / C_v$$

$$T = 0.287 * 10 \text{ ft} * 10 \text{ ft} / 0.29 \text{ ft}^2 / \text{day}$$

$$T = 98.9 \text{ days} \approx \mathbf{99 \text{ days}}$$

Correct answer is option B.

43. Problem

Correct Answer: (A) 0.9 cubic feet/ day / feet

Explanation:

Ref to NCEES PE handbook page 190 section 3.16.4 for flow nets

Permeability, $k = 0.0001$ ft/min

Number of flow channels, $N_f = 5$

Number of equipotential drops, $N_d = 12$

Head difference, $h = 15$ ft

From equation in page 191

Flow rate $Q = k \cdot h \cdot (N_f / N_d)$

$Q = 0.0001$ ft/ min * 15 ft * $(5/12)$

$Q = 0.000625$ cubic feet/min per feet width

Convert to per day

$1 \text{ day} = 24 \cdot 60 \text{ min} = 1440 \text{ min}$

$Q \text{ per day} = 0.000625 \cdot 1440 \text{ cubic feet/ day/feet}$

= 0.9 cubic feet/ day / feet

Correct answer is option A.

44. Problem

Correct Answer: (A) 0.35 ksi

Explanation:

Ref to ACI 318-14, page 316 Section 19.2.3.1,

Modulus of rupture

$$f_r = 7.5 \cdot \lambda \cdot \sqrt{f_c'}$$

For lightweight concrete from page 316, table 19.2.4.2,

$$\lambda = 0.75$$

$$f_r = 7.5 \cdot 0.75 \cdot \sqrt{4000}$$

$$f_r = 355.75 \text{ psi} = \mathbf{0.35 \text{ ksi}}$$

Correct answer is option A.

45. Problem

Correct Answer: (C) 1.75 in

Explanation:

Ref to ACI 318 -14, page 401 section 24.2.4.1.1

Calculate p' that is the ratio of compression rebar to $b*d$

$$b = 15 \text{ in}$$

$$d = 24 \text{ in} - 2 \text{ in} = 22 \text{ in}$$

Area of compression rebar

From NCEES PE handbook page 260, area of #5 bar = 0.31 in^2

$$A_s = 3 * 0.31 \text{ in}^2 = 0.93 \text{ in}^2$$

$$p' = 0.93 \text{ in}^2 / (15 \text{ in} * 22 \text{ in})$$

$$p' = 0.0028$$

From ACI 318-14 page 402, table 24.2.4.1.3

$\xi = 2.0$ for more than 60 months

From page 401, equation 24.2.4.1.1

$$\lambda = \xi / (1 + 50 * p')$$

$$\lambda = 2 / (1 + 50 * 0.0028)$$

$$\lambda = 1.754$$

Long term deflection = λ * short term deflection

$$= 1.754 * 1 \text{ in} = 1.754 \text{ in}$$

Hence the correct answer is Option C, **1.75 in**.

46. Problem

Correct Answer: (B)

Explanation:

A is the proportionality limit for which the linear relationship is valid.

B is the elastic limit and C is the yield point. For steel B and C are very close to each other.

D is the ultimate strength and E is the fracture point.

Hence Option B is the correct answer.

47. Problem

Correct Answer: (C) 1.83

Explanation:

Per AISC Steel manual 15th edition, AISC 360-16, page 16.1-46 section F1(c)

$$C_b = 12.5 * M_{max} / (2.5 M_{max} + 3M_a + 4 M_b + 3M_c)$$

$$M_{max} = 25 \text{ kip*ft} \quad M_a = 20 \text{ kip*ft} \quad M_b = 0 \quad M_c = 16 \text{ kip*ft}$$

$$C_b = 12.5 * 25 / (2.5 * 25 + 3 * 20 + 4 * 0 + 3 * 16)$$

$$C_b = 1.83 \quad \text{Correct answer is option C.}$$

48. Problem

Correct Answer: (C)

Explanation:

Knots are structural defects in wood. While it has minor impacts on density, stiffness and compressive strength, the tensile strength perpendicular to the grains is significantly weakened and risks of splitting increases. Hence the correct answer is Option C.

49. Problem

Correct Answer: (D) 1258 psi

Explanation:

Moisture Less than or equal to 19% is dry condition for lumber.

Ref to NDS 2018 Supplement page 33 Table 4A

For Alaska Yellow cedar No. 2, $F_b = 800 \text{ psi}$ (All adjustment values below are for F_b)

All adjustment values below are for F_b

From NDS 2018 page 11, Table 2.3.2, For wind load, $C_d = 1.6$

From NDS 2018 page 11, Table 2.3.3, For 110°F Dry, $C_t = 0.8$

From NDS supplement page 32, $C_m = 1$ for moisture less than 19% (dry)

From page 32, Size factor for 8 in width and No. 2 with thickness 4 in, $C_f = 1.3$

From page 32, Flat use factor for 8 in width and thickness 4 in, $C_{fu} = 1.05$

In absence of specific usage and spacing of beam, Repetitive member factor $C_r = 1$

From NDS 2018 page 29, Table 4.3.1,

$$F'_b = F_b * C_d * C_m * C_t * C_L * C_f * C_{fu} * C_i * C_r$$

$$F'_b = 800 \text{ psi} * 1.6 * 1 * 0.8 * 0.9 * 1.3 * 1.05 * 1 * 1 = 1257.98 \text{ psi}$$

Correct answer is option D, **1258 psi**.

50. Problem

Correct Answer: (C) 8 in

Explanation:

Ref to TMS 402/602-16, page C-213 section A 6.2.1,

For one story load bearing walls, minimum thickness shall be 6 in.

Ref to TMS 402/602-16, page C-212 Table A .5.1,

For fully grouted load bearing walls,

Maximum $h/t = 20$

$12 \text{ ft}/t = 20$

$t = 12 * 12 \text{ in} / 20$

$t = 144 \text{ in} / 20 = 7.2 \text{ in}$

Hence, the correct answer is option C, **8 in**.

51. Problem

Correct Answer: (D) Four #3

Explanation:

Ref to TMS 402/602-16 page C-56, Section 5.3.1.3,

Vertical reinforcement shall not be less than $0.0025 A_n$ and minimum 4 bars should be provided.

Minimum area required = $0.0025 * A_n$

= $0.0025 * 9.625 \text{ in} * 9.625 \text{ in}$

= 0.23 in^2

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #3 bars = 0.11 in^2

Area of #4 bars = 0.2 in^2

Although three #3 and two #4, satisfy the total area required, a minimum of four bars is needed.

Hence correct answer is Option D, **Four #3**.

52. Problem

Correct Answer: (D) 11

Explanation:

Hammer efficiency measured

$$= \text{Energy transferred} / \text{Potential energy} * 100\%$$

$$= (375 \text{ J} / 467 \text{ J}) * 100\%$$

$$= 80.3\%$$

Ref to NCEES PE handbook page 124, section 3.8.1.1.

$$N_{60} = (\text{Efficiency} / 60) * N$$

$$N_{60} = (80.3 / 60) * 8$$

$$N_{60} = 10.7.$$

Hence, the correct answer is Option D, **11**.

53. Problem

Correct Answer: (C) HSS 18.000 X 0.375

Explanation:

Max moment due to uniformly distributed = $w * L^2 / 8$

$$M = 3 \text{ kip/ft} * (25 \text{ ft})^2 / 8 = 234.37 \text{ kip*ft}$$

From AISC steel manual, page 3-141, Table 3-14,

For **HSS 18.000 X 0.375**

Allowable moment = 242 kip*ft

This is the lightest section that will satisfy the requirement.

HSS18.000x	0.500	328
	0.375^f	242
HSS16.000x	0.625	317
	0.500	257
	0.438	227
	0.375	194
	0.312 ^f	158
	0.250 ^f	123

Hence the correct answer is Option C.

54. Problem

Correct Answer: (D) 150%

Explanation:

Component of force acting along major axis

$$W_x = 1 \text{ kip/ft} * \cos(30^\circ) = 0.866 \text{ kip/ft}$$

$$M_x = W_x * L^2 / 8$$

$$M_x = 0.866 \text{ kip/ft} * (12\text{ft})^2 / 8 = 15.6 \text{ kip*ft}$$

Component of force acting along minor axis

$$W_y = 1 \text{ kip/ft} * \sin(30^\circ) = 0.5 \text{ kip/ft}$$

$$M_y = W_y * L^2 / 8$$

$$M_y = 0.5 \text{ kip/ft} * (12\text{ft})^2 / 8 = 9 \text{ kip*ft}$$

Total Utilization

$$= M_x/M_{xa} + M_y/M_{ya}$$

$$= 15.6/30 + 9/10 = 1.42 = 142\%$$

Hence the closest answer is Option D, **150%**.

55. Problem

Correct Answer: (B) 7.5 ft

Explanation:

$$F_y = 50 \text{ ksi}$$

$$E = 29000 \text{ ksi}$$

Calculate radius of gyration about weaker axis

$$r_y = \sqrt{I_y/A} = \sqrt{311.7 \text{ in}^4/68 \text{ in}^2} = 2.141 \text{ in}$$

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-48, equation F2-5

Limiting unbraced length for yielding

$$L_p = 1.76 * r_y * \sqrt{E/F_y}$$

$$L_p = 1.76 * 2.141 \text{ in} * \sqrt{29000 \text{ ksi}/50 \text{ ksi}}$$

$$L_p = 90.74 \text{ in} \approx \mathbf{7.5 \text{ ft}}$$

Hence the correct answer is Option B.

56. Problem

Correct Answer: (B) 9in

Explanation:

Shear depth, $d_v = 24$ in

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #4 bar, $A_v = 0.2$ in²

From TMS 402/602-16, page C-114, section 8.3.3.1,

Allowable stress in grade 60 rebar

$F_s = 32000$ psi

From TMS 402/602-16, page C-118, equation 8-27,

Shear stress carried by rebar with spacing 's'

$$F_{vs} = 0.5 * [(A_v * F_s * d_v) / (A_{nv} * s)]$$

$$s = 0.5 * [(A_v * F_s * d_v) / (A_{nv} * F_{vs})]$$

$$s = 0.5 * [(0.2 \text{ in}^2 * 32000 \text{ psi} * 24 \text{ in}) / (7.625 \text{ in} * 24 \text{ in} * 27.5 \text{ psi})] = 15.26 \text{ in}$$

From TMS 402/602-16, page C-119, section 8.3.5.2.1, spacing is limited to $d/2$

$$d/2 = 20 \text{ in} / 2 = 10 \text{ in}.$$

Hence the correct answer is Option B, **9in**.

57. Problem

Correct Answer: (C) 10.8 ksi

Explanation:

From NDS-2018 supplement page 34, Table 4A

For DFL, select structural, $E_{min} = 690000$ psi

For normal temperature, from NDS 2018, page 11, Table 2.3.3, $C_t = 1$

For wet condition, from NDS-2018 supplement, page 32, $C_m = 0.9$ for E_{min}

From NDS-2018 page 29, Table 4.3.1

$$E_{min}' = E_{min} * C_m * C_t * C_i * C_T$$

$$E_{min}' = 690000 \text{ psi} * 0.9 * 1 * 1 * 1 = 621000 \text{ psi}$$

From NDS -2018, page 17, section 3.3.3.8

$$F_{bE} = 1.2 * E_{min}' / R_b^2$$

$$F_{bE} = 1.2 * 621000 \text{ psi} / 8.3^2 = 10817.24 \text{ psi} = \mathbf{10.8 \text{ ksi}}$$

Correct answer is Option C.

58. Problem

Correct Answer: (C) 190 kip*ft

Explanation:

From NCEES PE handbook page 260, section 4.3.1,

Area of #6 bars = 0.44 in^2

Total $A_s = 5 * 0.44 \text{ in}^2 = 2.2 \text{ in}^2$

Assume that the depth of the compression block is less than 6in, i.e. lies inside the flange thickness.

From NCEES PE handbook page 264, section 4.3.2.2

$$a = (A_s * f_y) / (0.85 * f'_c * b_f)$$

$$a = (2.2 \text{ in}^2 * 60 \text{ ksi}) / (0.85 * 4 \text{ ksi} * 40 \text{ in})$$

$a = 0.971 \text{ in} < 6 \text{ in}$, assumption is valid.

$$M_n = A_s * f_y * (d - a/2)$$

$$M_n = 2.2 \text{ in}^2 * 60 \text{ ksi} * (18 \text{ in} - 0.971 \text{ in} / 2)$$

$$M_n = 2311.9 \text{ kip*in}$$

$$M_n = 192.7 \text{ kip*ft}$$

Closest answer is Option C , **190 kip*ft**.

59. Problem

Correct Answer: (D) 40 in

Explanation:

From AISC steel manual page 15th ed. Page 16.1-17, Table B4.1a, Case II

$$\text{Limiting } \lambda_r = 0.64 * \sqrt{(k_c * E / F_y)}$$

$$h = 12 \text{ in} - 2 * 1.5 \text{ in} = 9 \text{ in}$$

$$t = 1.5 \text{ in}, t_w = 0.75 \text{ in}$$

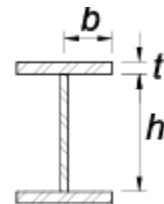
$$k_c = 4 / [\sqrt{(h/t_w)}] = 4 / [\sqrt{(9 \text{ in} / 0.75 \text{ in})}] = 1.155$$

Limiting value of $k_c = 0.76$

$$b/t = 0.64 * \sqrt{(k_c * E / F_y)} = 0.64 * \sqrt{(0.76 * 29000 \text{ ksi} / 50 \text{ ksi})}$$

$$b = 13.437 * t = 13.437 * 1.5 \text{ in} = 20.155 \text{ in}$$

Total permissible width = $2 * b = 40.3 \text{ in}$. Correct answer is Option D, **40 in**.



60. Problem

Correct Answer: (D) 1250 kips

Explanation:

$$KL = 15 \text{ ft} = 180 \text{ in} = L_c$$

$$F_y = 60 \text{ ksi}$$

From AISC steel manual page 15th ed. Page 1-20 and 1-21, Table 1-1

For W21X111

$$A = 32.6 \text{ in}^2$$

$$r_x = 9.05 \text{ in} \quad r_y = 2.9 \text{ in}$$

From steel manual page 16.1-36, eq. E3-4

$$F_e = \pi^2 * E / (L_c / r_y)^2$$

$$F_e = \pi^2 * 29000 \text{ ksi} / (180 \text{ in} / 2.9 \text{ in})^2 = 74.29 \text{ ksi}$$

From page 16.1-35, $F_y / F_e = 60 \text{ ksi} / 74.29 \text{ ksi} = 0.808 < 2.25$. Hence case (a) applies

$$F_{cr} = 0.658^{\frac{F_y}{F_e}} * F_y$$

$$F_{cr} = 0.658^{0.808} * 60 \text{ ksi} = 42.78 \text{ ksi} \quad \Phi_c = 0.9 \text{ from page 16.1-33 section E1}$$

$$\text{Design compressive strength} = \Phi_c * F_{cr} * A = 0.9 * 42.78 \text{ ksi} * 32.6 \text{ in}^2 = 1255.1 \text{ kips}$$

Correct answer is Option D, **1250 kips**.

61. Problem

Correct Answer: (B) 385 kips

Explanation:

Per TMS 402/602-16 page C-64, Table CC-6.1.2,

$$\text{Area of No. 8 bar, } A_s = 0.79 \text{ in}^2$$

$$A_{st} = 8 * 0.79 \text{ in}^2 = 6.32 \text{ in}^2$$

Per TMS 402/602-16 page C-114, section 8.3.3 (b)

$$F_s = 32000 \text{ psi for grade 60 rebar.}$$

Calculate radius of gyration

$$\text{For } 23.625 \text{ in} \times 23.625 \text{ in section, } I = b * h^3 / 12$$

$$I = 23.625 \text{ in} * (23.625 \text{ in})^3 / 12 = 25960.08 \text{ in}^4$$

$$A_n = 23.625 \text{ in} * 23.625 \text{ in} = 558.14 \text{ in}^2$$

$$r = \sqrt{I / A_n}$$

$$r = \sqrt{(25960.08 \text{ in}^4 / 558.14 \text{ in}^2)}$$

$$r = 6.82 \text{ in}$$

$$H/r = 20 \text{ ft} * 12(\text{in/ft}) / 6.82 \text{ in} = 35.2 < 99$$

From page C-115, section 8.3.4.2.1 (a), eq 8-18,

$$P_a = (0.25 * f'_m * A_n + 0.65 * A_{st} * F_s) * [1 - (H/140r)^2]$$

$$P_a = (0.25 * 2000 \text{ psi} * 558.14 \text{ in}^2 + 0.65 * 6.32 \text{ in}^2 * 32000 \text{ psi}) * [1 - (35.2/140)^2]$$

$$P_a = 384574.05 \text{ lbs}$$

$$P_a = 384.6 \text{ kips}$$

Hence the correct answer is Option B, **385 kips**.

62. Problem

Correct Answer: (C) Wall C

Explanation:

Wall A is a blocked shear wall as it has framing on all sides. Wall B, C and D are unblocked walls because framing is not present on all sides.

From AWC- SDPWS 2015, page 28, Table 4.3.4,

Max aspect ratio for blocked WSP walls = 3.5:1 ,

Wall A aspect ratio = $12/4 = 3$. O.K.

Max aspect ratio for unblocked WSP walls = 2:1

Wall B, C and D aspect ratios are all less than or equal to 2. O.K.

From AWC- SDPWS 2015, page 25, Section 4.3.3.2, the height of unblocked shear wall is limited to 16ft.

Hence **Wall C**, 18 ft tall ,does not meet the requirement and Option C is the answer.

63. Problem

Correct Answer: (A) 855 kips

Explanation:

Gross area, $A_g = 18 \text{ in} \times 18 \text{ in} = 324 \text{ in}^2$

$A_{st} = 8 \times 0.6 \text{ in}^2 = 4.8 \text{ in}^2$

From ACI 318-14, page 350, equation 22.4.2.2,

$P_o = 0.85 \cdot f'_c \cdot (A_g - A_{st}) + f_y \cdot A_{st}$

$P_o = 0.85 \cdot 5 \text{ ksi} \cdot (324 \text{ in}^2 - 4.8 \text{ in}^2) + 60 \text{ ksi} \cdot 4.8 \text{ in}^2 = 1644.6 \text{ kips}$

From ACI 318-14, page 350, Table 22.4.2.1,

$P_{n, \max} = 0.8 \cdot P_o = 0.8 \cdot 1644.6 \text{ kips} = 1315.7 \text{ kips}$

$\Phi \cdot P_{n, \max} = 0.65 \cdot 1315.7 \text{ kips} = 855.2 \text{ kips}$

Per ACI 318-14, page 350, section R22.4.2.1, the 0.8 factor applied to 'Po' is to account for eccentricity to depth ratios of 0.1 for tied columns.

Eccentricity considered = $0.1 \cdot 18 \text{ in} = 1.8 \text{ in}$

Eccentricity due to applied moment = $M / (\Phi \cdot P_{n, \max})$

$= 100 \text{ kip} \cdot \text{ft} / 855.2 \text{ kips} = 0.117 \text{ ft} = 1.4 \text{ in}$, this is less than 1.8 in

Thus axial load capacity is not compromised and capacity $\approx$ **855 kips**

Correct answer is Option A.

64. Problem

Correct Answer: (C) Install a tuned mass damper

Explanation:

Increasing the overall mass of the structure by beefing up superstructure and substructure elements is not an economical solution to the problem and may have consequences with seismic design. Increasing the height is incorrect and impractical. Reducing the stiffness of the bracing may alter the frequency but will also have severe consequences to lateral and gravity load resisting capacity. Tuned mass dampers are advanced devices that help mitigate vibrations caused due to wind and seismic loads. Hence Option C is the correct answer.

65. Problem

Correct Answer: (A) 5.19 in

Explanation:

From AISC Steel manual 15th edition, page 1-22, Table 1-1,

Area of W18X50, $A_1 = 14.7 \text{ in}^2$

Depth, $d = 18 \text{ in}$

Location of centroid of the section from the top of composite section

$$y_1 = 4.5 \text{ in} + 2.5 \text{ in} + 18 \text{ in} / 2 = 16 \text{ in}$$

Modular ratio, $n = 8$

Effective width of concrete, $b_{\text{eff}} = 8 \text{ ft} = 96 \text{ in}$

Effective width of transformed concrete, $b = 96 \text{ in} / n = 96 \text{ in} / 8 = 12 \text{ in}$

Area of transformed concrete, $A_2 = 4.5 \text{ in} * b = 4.5 \text{ in} * 12 \text{ in} = 54 \text{ in}^2$

Distance of centroid of transformed concrete slab from the top of composite section,

$$y_2 = 4.5 \text{ in} / 2 = 2.25 \text{ in}$$

Taking first moment of area from the top, distance of centroid of composite section from top of section, $y = (A_1 * y_1 + A_2 * y_2) / (A_1 + A_2)$

$$y = (14.7 \text{ in}^2 * 16 \text{ in} + 54 \text{ in}^2 * 2.25 \text{ in}) / (14.7 \text{ in}^2 + 54 \text{ in}^2) = \mathbf{5.19 \text{ in}}$$

Correct answer is Option A.

66. Problem

Correct Answer: (C) 515 kips

Explanation:

From AISC steel manual 15th ed., page 3-14, Fig. 3-3, the distance between top of steel and location of effective concrete flange force is Y_2

$$Y_2 = 4 \text{ in}$$

$$\text{Total factored moment} = 240 \text{ kip}\cdot\text{ft} + 400 \text{ kip}\cdot\text{ft} = 640 \text{ kip}\cdot\text{ft}$$

From AISC steel manual 15th ed., page 3-174 and 3-175, Table 3-19

For $Y_2 = 4 \text{ in}$ and W18X50 with LRFD moment capacity $650 \text{ kip}\cdot\text{ft}$,

$$\Sigma Q_n = 521 \text{ kips}$$

Closest answer is Option C, **515 kips**.

67. Problem

Correct Answer: (C) 2 L 2X2X5/16

Explanation:

Span of the truss, $L = 20$ ft

Net service load from Case A

$= 5 \text{ kip} + 15 \text{ kip} = 20 \text{ kip}$ Tension

Net service load from Case B

$= 5 \text{ kip} - 18 \text{ kip} = 13 \text{ kip}$ compression

Unbraced length $= 8$ ft

Ref to AISC steel manual 15th ed. Page 4-149, Table 4-8

For unbraced length 8ft, Allowable Compression for 2 L 2X2X5/16 $= 13.5$ kips, O.K.

Ref to AISC steel manual 15th ed. Page 5-51, Table 5-8

For 2 L 2X2X5/16, minimum tensile strength $= 50$ kips, O.K.

Hence the correct answer is Option C.

68. Problem

Correct Answer: (C) 45 in

Explanation:

Size of weld, $w = 0.25$ in

Length of weld, $l = 70$ in $> 100 \cdot w$

From AISC steel manual 15th ed. , page 16.1-120, eqn. J2-1

$$\beta = 1.2 - 0.002 \cdot (l/w)$$

$$\beta = 1.2 - 0.002 \cdot (70 \text{ in} / 0.25 \text{ in})$$

$$\beta = 0.64$$

Check, $300 \cdot w = 300 \cdot 0.25 \text{ in} = 75 \text{ in}$, i.e. length does not exceed $300 \cdot w$

Hence effective length $= \beta \cdot 70 \text{ in} = 0.64 \cdot 70 \text{ in} = 44.8 \text{ in} \approx \mathbf{45 \text{ in}}$

Correct answer is Option C.

69. Problem

Correct Answer: (C) 0.93

Explanation:

The shear lag factor for W section with plates at top and bottom is calculated by splitting the W section into two equivalent half T sections

From AISC steel manual 15th ed, page 1-28, Table 1-1, for W8X40

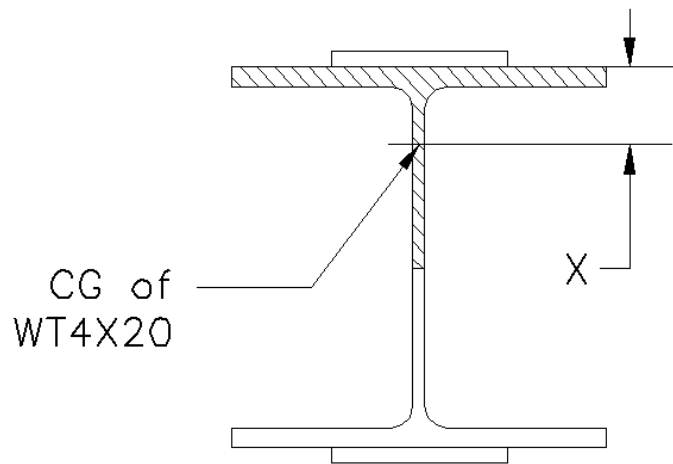
Depth = 8.25 in and weight = 40 lbs/ft

The half T section depth should be 4.125in and weight should be 20 lb/ft

WT 4 X 20 matches the criteria

From AISC steel manual 15th ed, page 1-70 and 1-71, Table 1-8, for WT 4X20

Distance from centroid of the T section to top of the flange = 0.735 in



Thus connection eccentricity, $X = 0.735$ in

Length of connection, $L = 10$ in

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$$U = 1 - x/l$$

$$= 1 - 0.735 \text{ in} / 10 \text{ in}$$

$$= \mathbf{0.93}$$

Correct answer is Option C.

70. Problem

Correct Answer: (B) 3/8 in

Explanation:

The connection is in double shear. So the splice plates share half of the loads each.

$$P_u = 100 \text{ kips} / 2 = 50 \text{ kips}$$

Width of plate, $b = 6 \text{ in}$ thickness, $t = ?$

$$F_y = 36 \text{ ksi} \quad F_u = 58 \text{ ksi}$$

From AISC steel manual 15th ed. Page 16.1-137, Section J4.1

For tensile yielding,

$$P_u = \phi_y F_y A_g$$

$$50 \text{ kips} = 0.9 * 36 \text{ ksi} * 6 \text{ in} * t$$

$$t = 0.257 \text{ in} > 1/4 \text{ in, try with } t = 3/8 \text{ inches}$$

Per AISC steel manual 15th ed. page 16.1-20, section B4.3b,

$$\text{hole dimension for net area} = 13/16 \text{ in} + 1/16 \text{ in} = 0.875 \text{ in}$$

$$\text{Net area using thickness} = (6 \text{ in} - 2 * 0.875 \text{ in}) * 3/8 \text{ in} = 1.59 \text{ in}^2$$

$$\text{Design rupture strength} = \phi_r F_u A_n = 0.75 * 58 \text{ ksi} * 1.59 \text{ in}^2 = 69 \text{ kips, O.K.}$$

Hence the correct answer is Option B, **3/8 in.**

71. Problem

Correct Answer: (A) 134.4 kips

Explanation:

From AISC steel manual 15th ed, page 1-47, Table 1-7, for L 5X5X1/2

Distance from centroid of the base of leg, $x = 1.42 \text{ in}$

$$\text{Gross area of section, } A_g = 4.79 \text{ in}^2$$

Length of connection in the direction of tension, $L = 4 \text{ in}$

From AISC steel manual 15th ed, page 16.1-30, Table D3.1, Case 2

$$U = 1 - x/L$$

$$U = 1 - 1.42 \text{ in} / 4 \text{ in} = 0.645$$

For welded connections, no hole reductions for net area

From AISC steel manual 15th ed page 16.1-29, Eqn. D3-1, $A_e = A_n * U$

$$A_e = 0.645 * 4.79 \text{ in}^2 = 3.09 \text{ in}^2.$$

From AISC steel manual 15th ed page 16.1-28, section D2,

For tensile yielding,

$$\Phi * F_y * A_g = 0.9 * 36 \text{ ksi} * 4.79 \text{ in}^2 = 155.2 \text{ kips}$$

For tensile rupture,

$$\Phi * F_u * A_e = 0.75 * 58 \text{ ksi} * 3.09 \text{ in}^2 = \mathbf{134.4 \text{ kips}}$$

Correct answer is Option A.

72. Problem

Correct Answer: (D) 41.2 kips

Explanation:

Length of welds (single line), $L = 6 \text{ in}$

Total vertical load in connection = P

Total moment in the connection = $P * 3 \text{ in}$

From AISC steel manual 15th ed. Page 8-14, Fig 8-6,

For single line vertical weld, $I = L^3 / 12$

For two lines, $I = 2 * L^3 / 12 = 2 * (6 \text{ in})^3 / 12 = 36 \text{ in}^3$

Section modulus, $S = I / (L/2) = 36 \text{ in}^3 / 3 \text{ in} = 12 \text{ in}^2$

Shear per unit length of weld, $V = P / (2 * L) = P / 12 = 0.0833 P$

Max tension in the weld, $T = \text{Moment} / S = P * 3 / 12 = P / 4 = 0.25 P$

Resultant per unit length = $\sqrt{(V^2 + T^2)} = \sqrt{[(0.0833 P)^2 + (0.25 P)^2]} = 0.2635 P$

Angle between weld line and resultant, $\Theta = \tan^{-1} (T/V) = \tan^{-1} (0.25 P / 0.0833 P) = 71.6^\circ$

From AISC steel manual 15th ed. Page 8-9, effect of load angle = $(1 + 0.5 \sin^{1.5} \Theta)$

Size of 1/2 inch weld in 16th of an inch, $1/2 \text{ in} = 8/16 \text{ in}$. Hence $D = 8$

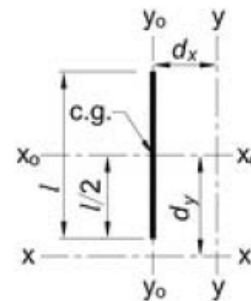
From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, $R_n / \Omega = 0.928 * D * l$

Resultant = Allowable stress per unit length of weld including load angle effect

Resultant = $0.928 * D * (1 + 0.5 \sin^{1.5} \Theta)$

$0.2635 P = 0.928 * 8 * (1 + 0.5 * \sin^{1.5} 71.6^\circ)$

$P = \mathbf{41.2 \text{ kips}}$ Correct answer is Option D.



73. Problem

Correct Answer: (C) 3/4 in.

Explanation:

Eccentricity, $e = 9$ in,

No. of bolts in one row, $n = 4$

From AISC steel manual 15th edition, page 7-43, Table 7-8,

For $s = 3$ in, $e = 9$ in, $n = 4$, we get, $C = 3.01$

Allowable Strength of individual bolt = r_n / Ω

Available strength of the connection, $P_{allowable} = C * r_n / \Omega = 20$ kips

$r_n / \Omega = 20 \text{ kips} / C$

$r_n / \Omega = 20 \text{ kips} / 3.01 = 6.64$ kips

From AISC steel manual 15th edition, page 7-25, Table 7-3,

Capacity of 5/8 in. diameter bolt for STD holes in single shear = 5.42 kips

Capacity of 3/4 in. diameter bolt for STD holes in single shear = 7.91 kips

Hence the correct answer is Option C, **3/4 in.**

74. Problem

Correct Answer: (C) 4.83 kips

Explanation:

No. of bolts, $n_b = 1$

From AISC steel manual 15th edition, page 16.1-134, section J3.8, $D_u = 1.13$

Tension in the bolt, $T_a = 5$ kips

From AISC steel manual 15th edition, page 7-24, Table 7-3, For 3/4 in. diameter bolts

Allowable shear resistance, $V_r = 6.33$ kips (STD hole)

Min. Bolt pretension, $T_b = 28$ kips

From AISC steel manual 15th edition, page 16.1-135, eqn. J3-5b

$K_{sc} = 1 - 1.5 * T_a / (D_u * T_b * n_b)$

$K_{sc} = 1 - 1.5 * 5 \text{ kips} / (1.13 * 28 \text{ kips} * 1) = 0.763$

Available shear strength due to interaction = $K_{sc} * V_r = 0.763 * 6.33 \text{ kips} = \mathbf{4.83 \text{ kips}}$

Correct answer is Option C.

75. Problem

Correct Answer: (C) 20

Explanation:

From NDS 2018 page 182, Table L4

Diameter of 16d common nails = 0.162 in Length = 3.5 in

From NDS 2018 page 112, Table 12P

For 12 gauge thick steel member into DFL member with 16d common nails,

$Z = 149 \text{ lbs}$ (All adjustment factors = 1)

Required penetration = $10 \cdot D = 10 \cdot 0.162 \text{ in} = 1.62 \text{ in}$ (enough for 4X8 member)

Required number of nails = $2800 \text{ lbs} / Z = 2800 \text{ lbs} / 149 \text{ lbs} = 18.79$

Correct answer is Option C, **20**.

76. Problem

Correct Answer: (C) 4.44 ksf

Explanation:

Length of footing, $L = 5 \text{ ft}$

Breadth of footing, $B = 5 \text{ ft}$

Axial load, $P = 50 \text{ kips}$

Moment, $M = 50 \text{ kip} \cdot \text{ft}$

Eccentricity, $e = M/P$

$e = 50 \text{ kip} \cdot \text{ft} / 50 \text{ kip} = 1 \text{ ft} = 12 \text{ in}$

Check $B/6 = 5 \text{ ft} / 6 = 60 \text{ in} / 6 = 10 \text{ in}$

Thus $e > B/6$

From NCEES PE handbook page 93,

$Q_{\max} = 4P/[3 \cdot L \cdot (B - 2 \cdot e)]$

$Q_{\max} = 4 \cdot 50 \text{ kips} / [3 \cdot 5 \text{ ft} \cdot (5 \text{ ft} - 2 \cdot 1 \text{ ft})]$

$Q_{\max} = \mathbf{4.44 \text{ ksf}}$

Correct answer is Option C.

77. Problem

Correct Answer: (C) 3.25 ft

Explanation:

Total Axial load, $P = 100 \text{ kips} + 150 \text{ kips} = 250 \text{ kips}$

Total Moment, $M = 50 \text{ kip}\cdot\text{ft} + 80 \text{ kip}\cdot\text{ft} = 130 \text{ kip}\cdot\text{ft}$

Eccentricity, $e = M/P = 130 \text{ kip}\cdot\text{ft} / 250 \text{ kips} = 0.52 \text{ ft}$

For entire footing area to be effective in bearing the loads,

$$e \leq B/6$$

$$B \geq 6 \cdot e$$

$$B \geq 6 \cdot 0.52 \text{ ft}$$

$$B \geq 3.12 \text{ ft}$$

Correct answer is option C, **3.25 ft**.

78. Problem

Correct Answer: (C) #6 @ 12 in. c/c

Explanation:

Depth, $D = 18 \text{ in}$

Breadth, $B = 1 \text{ ft} = 12 \text{ in}$ (rebar per ft of width calculated)

From ACI 318-14, page 104, Table 8.6.1.1

For $f_y = 60000 \text{ psi}$

$$A_{smin} = 0.0018 \cdot A_g$$

$$A_{smin} = 0.0018 \cdot D \cdot B$$

$$A_{smin} = 0.0018 \cdot 18 \text{ in} \cdot 12 \text{ in} = 0.389 \text{ in}^2$$

From ACI 318-14, page 507, Appendix A

$$\text{Area of \#4 bar} = 0.2 \text{ in}^2$$

$$\text{Area of \#5 bar} = 0.31 \text{ in}^2$$

$$\text{Area of \#6 bar} = 0.44 \text{ in}^2 \text{ (satisfies requirement)}$$

Hence the correct answer is Option C, **#6 @ 12 in. c/c**.

79. Problem

Correct Answer: (B) 19 ton

Explanation:

Weight of hammer, $W = 700$ lbs

Height of drop, $h = 12$ ft

Driving energy, $E = W \cdot h$

$E = 700 \text{ lbs} \cdot 12 \text{ ft} = 8400 \text{ ft-lbs}$

Penetration of pile per hammer blow, $S = 1 \text{ inch} / 3 = 0.333 \text{ in}$

From NCEES PE handbook, page 59, section 2.3.4.2

$P = 2 \cdot E / (S + k)$ [$k = 0.1$ for single acting air hammer]

$P = 2 \cdot 8400 \text{ lbs} / (0.333 + 0.1) = 38799.08 \text{ lbs}$

From NCEES PE handbook page 3,

$2000 \text{ lbs} = 1 \text{ ton}$

$38799.08 \text{ lbs} \approx \mathbf{19 \text{ ton}}$

Correct answer is option B.

80. Problem

Correct Answer: (B) $0.583 \text{ in}^2/\text{ft}$

Explanation:

$d = 18 \text{ in} - 3.5 \text{ in} = 14.5 \text{ in}$

$M_u = 13583 \text{ lb} \cdot \text{ft}/\text{ft} = 13583 \cdot (12/1000) \text{ kip} \cdot \text{in}/\text{ft}$

$M_u = 163 \text{ kip} \cdot \text{in}/\text{ft}$

Calculate area of steel for 12 in width

From NCEES PE handbook page 264, section 4.3.2.2

$a = A_s \cdot f_y / (0.85 \cdot f'_c \cdot b)$

$a = A_s \cdot 60 / (0.85 \cdot 4.5 \cdot 12)$

$a = 1.307 \cdot A_s$

Assume tension controlled section, From ACI 318-14, page 343, Table 21.2.3

$\Phi = 0.9$

From NCEES PE handbook page 264, section 4.3.2.2

$M_n = A_s \cdot f_y \cdot (d - a/2)$

$$M_n = A_s \cdot 60 \cdot (14.5 - 1.307 \cdot A_s / 2)$$

$$M_n = 870 \cdot A_s - 39.21 \cdot A_s^2$$

$$M_u = \Phi \cdot M_n$$

$$163 = 0.9 \cdot (870 \cdot A_s - 39.21 \cdot A_s^2)$$

$$35.29 \cdot A_s^2 - 783 \cdot A_s + 163 = 0$$

$$\text{Solving, we get } A_s = 0.21 \text{ in}^2$$

For flexure at cantilevered heel, the minimum flexural reinforcement limit is determined from ACI 318-14, page 137, section 9.6.1.2, because of lack of redundancy (similar to statically determinate beam).

Minimum reinforcement should be greater of

$$\text{a) } 3(\sqrt{f'_c}) \cdot b_w \cdot d / f_y = 3(\sqrt{4500}) \cdot 12 \text{ in} \cdot 14.5 \text{ in} / 60000 = 0.583 \text{ in}^2$$

$$\text{b) } 200 \cdot b_w \cdot d / f_y = 200 \cdot 12 \text{ in} \cdot 14.5 / 60000 = 0.58 \text{ in}^2$$

Hence minimum requirement controls and required reinforcement = **0.583 in²/ft**

Correct answer is option B.